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(54) **METHODS AND APPARATUS TO ESTIMATE AUDIENCE SIZES OF MEDIA USING DEDUPLICATION BASED ON MULTIPLE VECTORS OF COUNTS**

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(52) **U.S. Cl.**

CPC **G06F 16/215** (2019.01); **G06Q 30/0205** (2013.01); **H04N 21/4667** (2013.01); **G06Q 50/265** (2013.01)

(58) **Field of Classification Search**

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See application file for complete search history.

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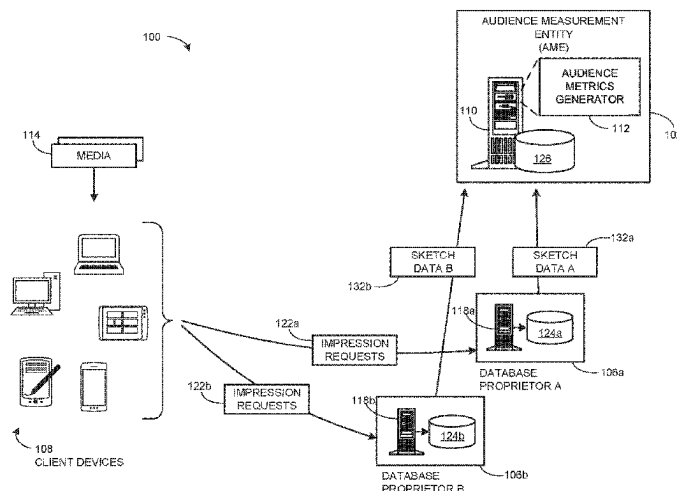
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(57)

ABSTRACT

Disclosed examples to estimate audience sizes of media include a coefficient generator to determine coefficient values for a polynomial based on normalized weighted sums of variances, a normalized weighted sum of covariances, and cardinalities corresponding to a first plurality of vectors of counts from a first database proprietor and a second plurality of vectors of counts from a second database proprietor, a real roots solver to determine a real root value of the polynomial, the real root value indicative of a number of audience members represented in the first plurality of vectors of counts that are also represented in the second plurality of vectors of counts, and an audience size generator to determine the audience size based on the real root value and the cardinalities of the first plurality of vectors of counts and the second plurality of vectors of counts.

14 Claims, 6 Drawing Sheets



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H04N 21/466 (2011.01)
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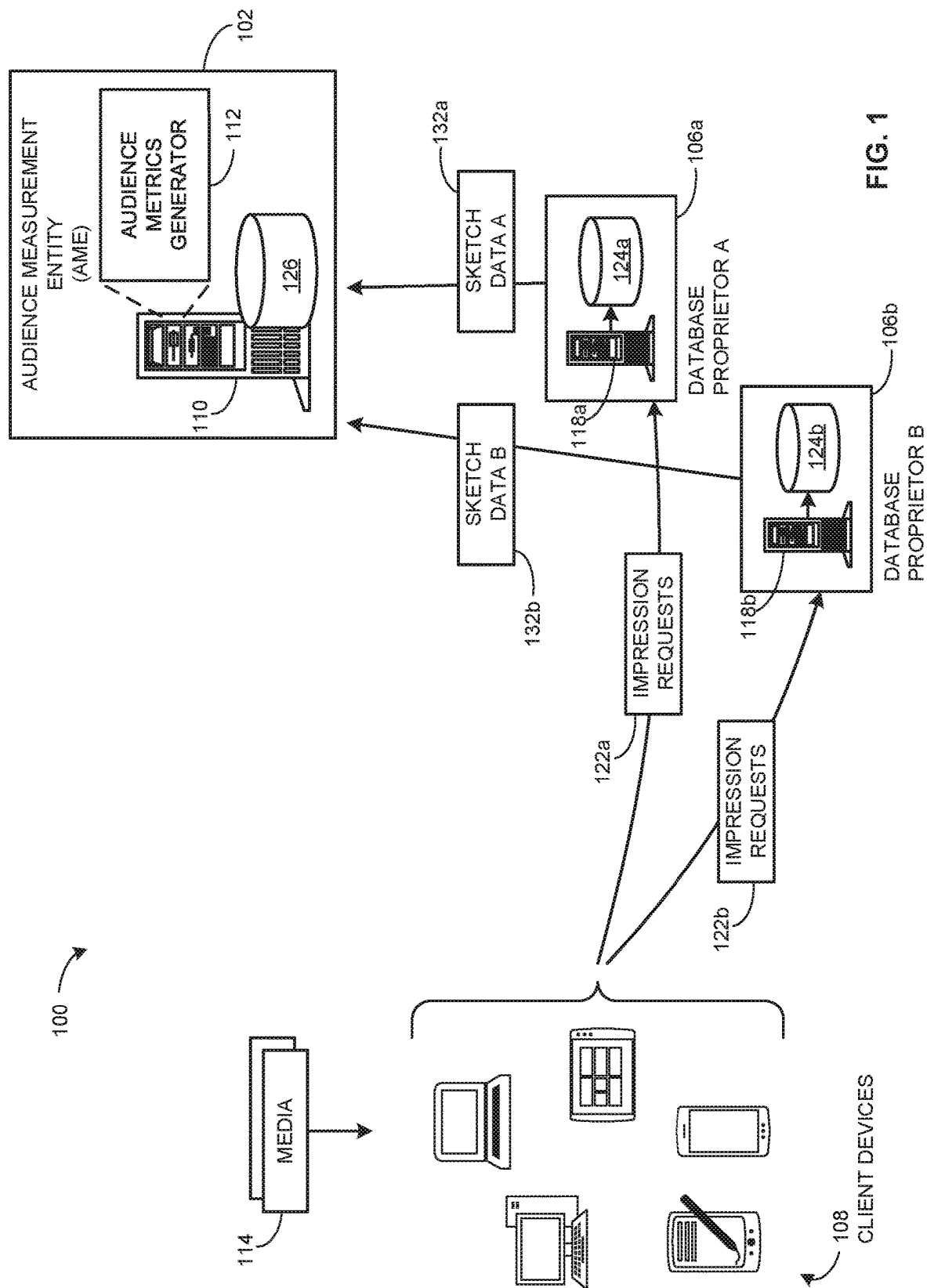
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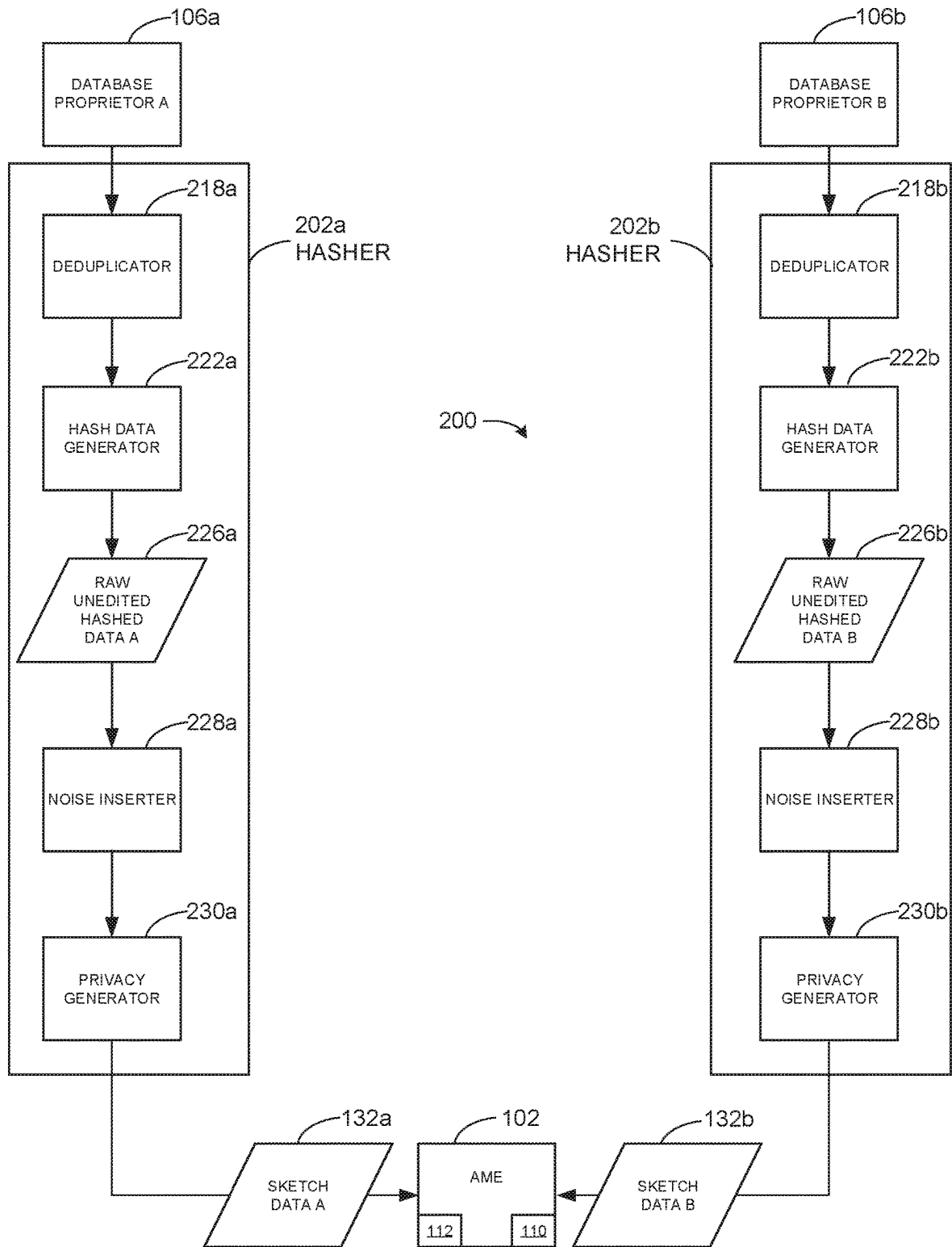


FIG. 2

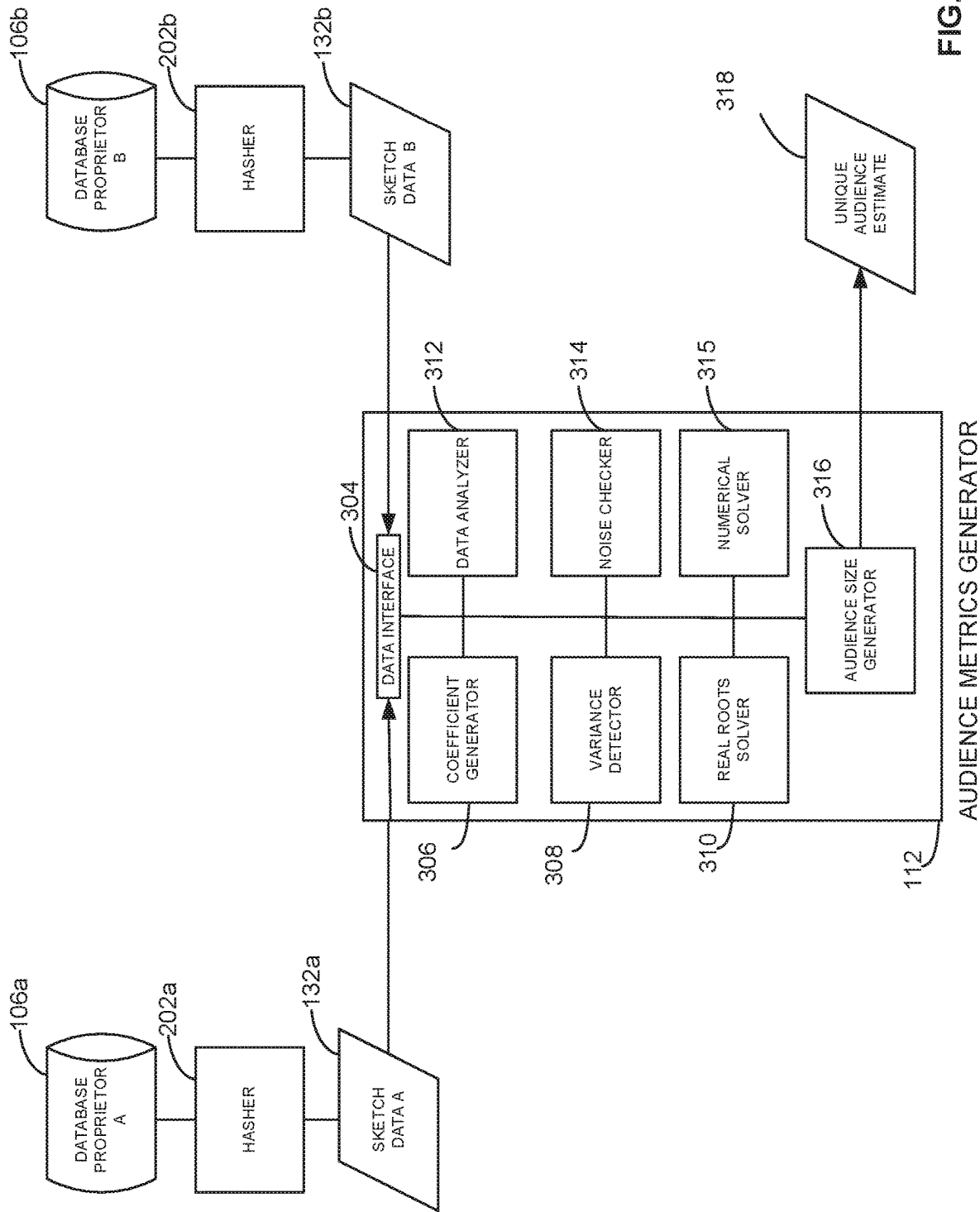


FIG. 3

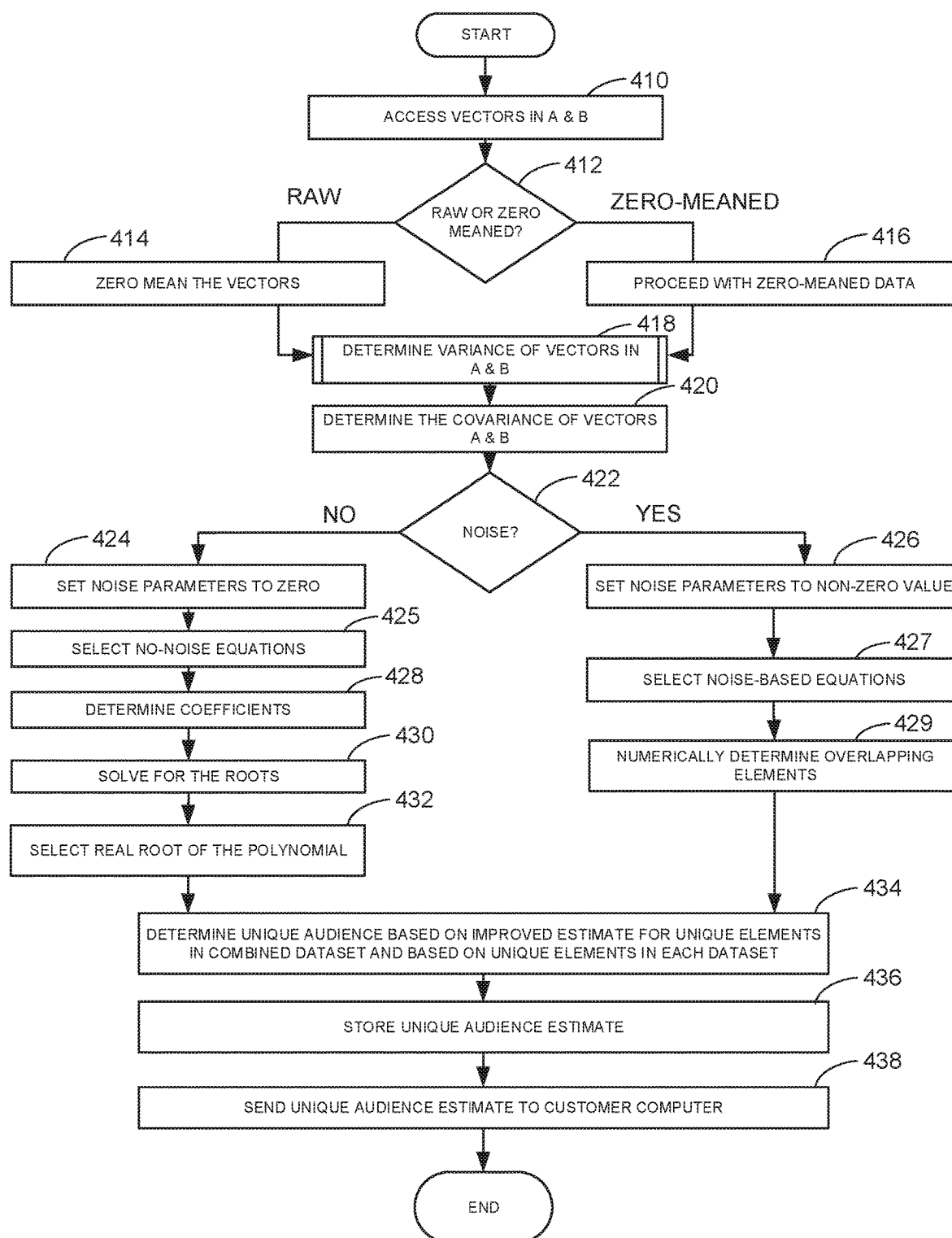


FIG. 4

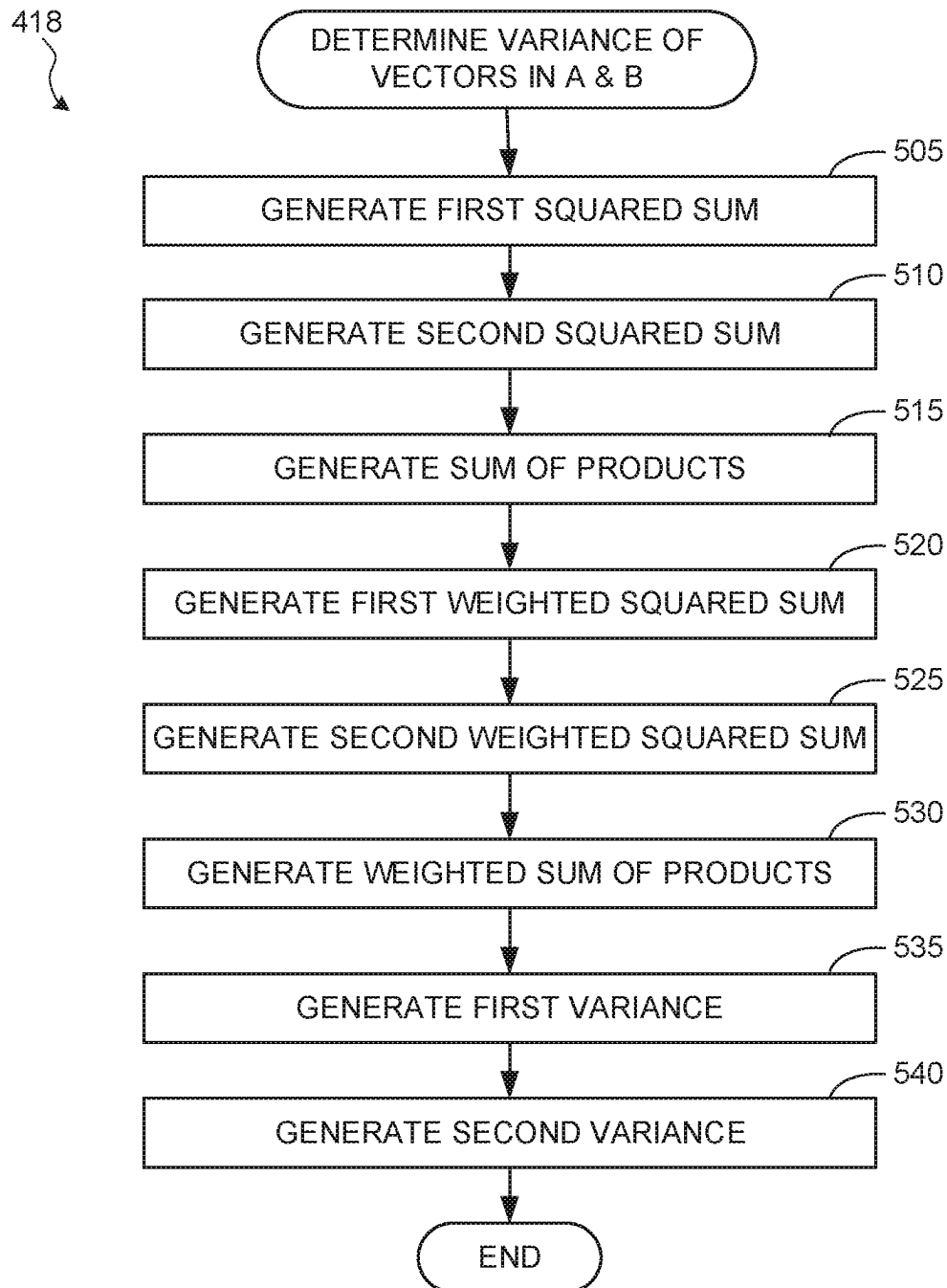


FIG. 5

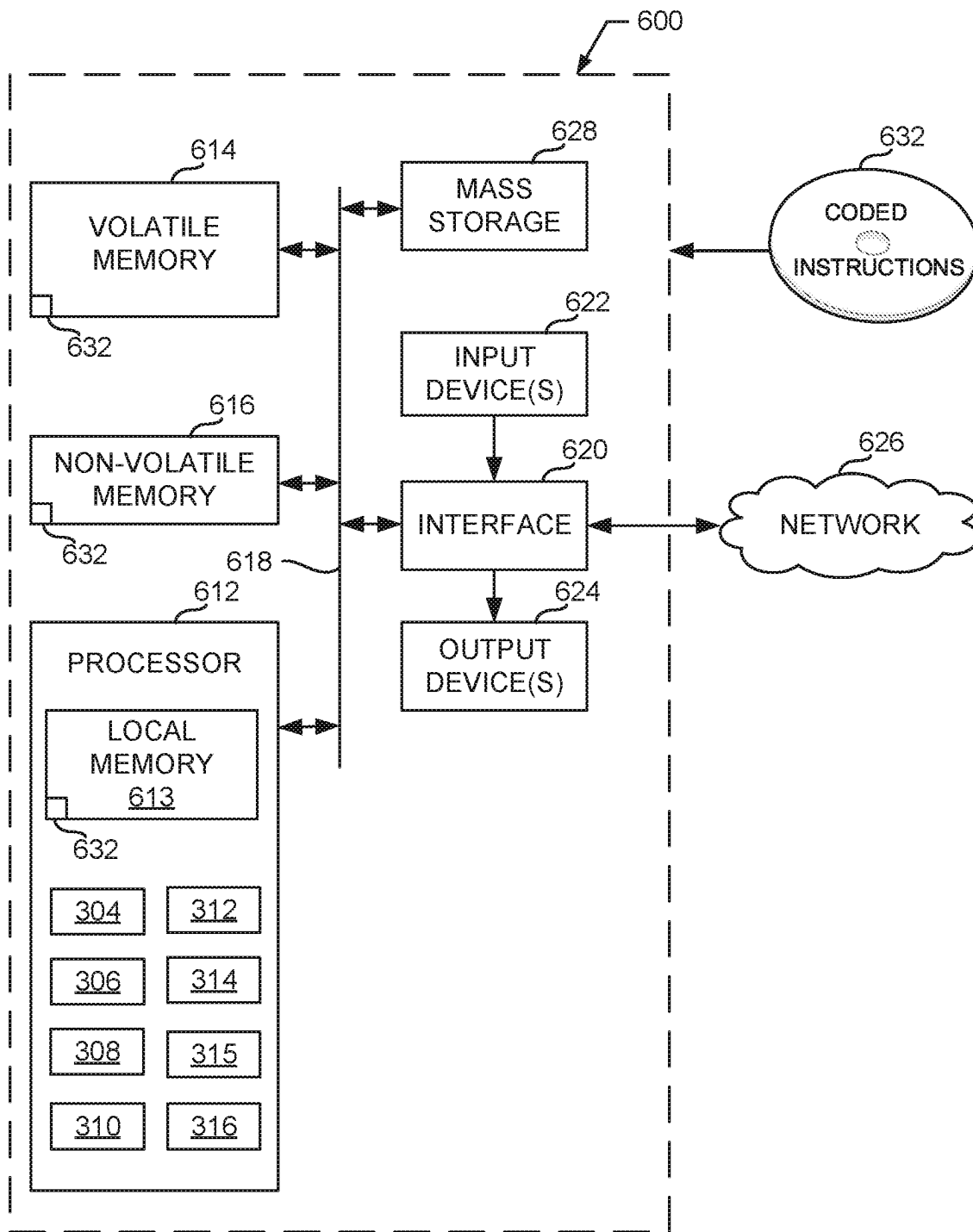


FIG. 6

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METHODS AND APPARATUS TO ESTIMATE AUDIENCE SIZES OF MEDIA USING DEDUPLICATION BASED ON MULTIPLE VECTORS OF COUNTS

CROSS-REFERENCE TO RELATED APPLICATIONS

This disclosure is a continuation of U.S. patent application Ser. No. 16/917,459, filed Jun. 30, 2020, issued as U.S. Pat. No. 12,032,535, the entirety of which is hereby incorporated by reference.

FIELD OF THE DISCLOSURE

This disclosure relates generally to computer-based audience monitoring and, more particularly, to methods and apparatus to estimate audience sizes of media using deduplication based on multiple vectors of counts.

BACKGROUND

Monitoring accesses to internet-based media can be accomplished using cookies. Cookies are set on computing devices by web servers to identify those computing devices during subsequent visits to those web servers. Two types of cookies include first-party cookies and third-party cookies. A first-party cookie is a cookie set by and accessible by a web server that provides web pages and/or media to a computing device. A third-party cookie is set by a third-party server separate from the web server that provides the web pages and/or media to the computing device. A first-party cookie allows a web server to monitor media accesses, recall user preferences, and/or deliver customized web page experiences to the same user over multiple sessions with the web server. A third-party cookie is typically used solely for collecting media access information to, for example, generate audience metrics information across the Internet and/or to serve Internet-based advertising from ad servers. Thus, while first-party cookies limit media access monitoring capabilities to only first-party web servers that actually serve the media, third-party cookies make it possible for third parties to monitor media accesses without third parties needing to participate in serving media or hosting web pages across the Internet to computing devices.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is an example network-based system for logging impressions of accesses to server-based media.

FIG. 2 is a sketch data generation system to facilitate database proprietors of FIG. 1 to provide sketch data to the audience measurement entity of FIG. 1 for use in estimating audience sizes of media.

FIG. 3 is a block diagram of an example audience metrics generator of FIGS. 1 and 2 to estimate audience sizes of media using deduplication based on multiple vectors of counts in accordance with teachings of this disclosure.

FIG. 4 is a flowchart representative of example machine-readable instructions which may be executed to implement the example audience metrics generator of FIGS. 1-3 to estimate audience sizes of media using deduplication based on multiple vectors of counts in accordance with teachings of this disclosure.

FIG. 5 is a flowchart representative of example machine-readable instructions which may be executed to implement the example audience metrics generator of FIGS. 1-3 to

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determine a variance of server-collected impression data in accordance with teachings of this disclosure.

FIG. 6 is a block diagram of an example processing platform structured to execute the instructions of FIGS. 4 and 5 to implement the example audience metrics generator of FIGS. 1-3 to estimate audience sizes of media using deduplication based on multiple vectors of counts in accordance with teachings of this disclosure.

The figures are not to scale. In general, the same reference numbers will be used throughout the drawing(s) and accompanying written description to refer to the same or like parts. Connection references (e.g., attached, coupled, connected, and joined) are to be construed broadly and may include intermediate members between a collection of elements and relative movement between elements unless otherwise indicated. As such, connection references do not necessarily infer that two elements are directly connected and in fixed relation to each other.

Descriptors “first,” “second,” “third,” etc. are used herein when identifying multiple elements or components which may be referred to separately. Unless otherwise specified or understood based on their context of use, such descriptors are not intended to impute any meaning of priority, physical order or arrangement in a list, or ordering in time but are merely used as labels for referring to multiple elements or components separately for ease of understanding the disclosed examples. In some examples, the descriptor “first” may be used to refer to an element in the detailed description, while the same element may be referred to in a claim with a different descriptor such as “second” or “third.” In such instances, it should be understood that such descriptors are used merely for ease of referencing multiple elements or components.

DETAILED DESCRIPTION

There are many database proprietors operating on the Internet. These database proprietors provide services to large numbers of subscribers. In exchange for the provision of services, the subscribers register with the database proprietors and provide personally identifiable information (PII). The types of information collected as part of PII can differ between database proprietors but typically include demographic information (e.g., gender, age, geographic location, income, occupation, etc.), names, email addresses, phone numbers, and/or other information that could be used to identify a particular individual. Examples of database proprietors include social network sites (e.g., Facebook, Twitter, MySpace, etc.), multi-service sites (e.g., Yahoo!, Google, Axiom, Catalina, etc.), online retailer sites (e.g., Amazon.com, Buy.com, etc.), credit reporting sites (e.g., Experian), streaming media sites (e.g., YouTube, Hulu, etc.), etc. These database proprietors set cookies (e.g., first-party cookies) and/or other device/user identifiers on the client devices of their subscribers to enable the database proprietors to recognize those subscribers visiting their web sites.

Techniques for deduplication of subscribers between two database proprietors are used to estimate the unique audience size (e.g., number of unique subscribers) between the two database proprietors. The unique audience size is the number of audience members (e.g., subscribers) subscribed to at least one of the two database proprietors and that accessed media served by respective servers (e.g., web servers, media servers, ad servers, etc.) of the database proprietors. That is, an audience member subscribed to both of the database proprietors is only counted once in the unique audience size of the media item. Due to privacy

concerns, subscribers of the first database proprietor are known to the first database proprietor, but not to the second database proprietor. Similarly, subscribers of the second database proprietor are known to the second database proprietor, but not to the first database proprietor.

Examples disclosed herein may be used by a monitoring entity to estimate the unique audience size of media served by different database proprietors without compromising the privacies of the subscribers of each database proprietor (e.g., without sharing subscribers' PII outside a database proprietor). In some examples, the monitoring entity can be one of the database proprietors which becomes a third-party entity relative to the other database proprietors. Alternatively, the monitoring entity can be a third-party audience measurement entity (AME) that is a trusted (e.g., neutral) third party for providing accurate usage statistics (e.g., The Nielsen Company, LLC).

The unique audience size can be estimated for a particular subset of subscribers of each database proprietor. For example, each database proprietor can filter its subscribers to create a subset which includes only those subscribers who have accessed a particular media that is of interest for audience metrics analysis. As such, the monitoring entity can deduplicate between the subsets from each database proprietor to estimate the number of unique audience members who have accessed the particular media. Additionally, each database proprietor can further filter the subset based on subscribers that accessed the media and that correspond to a particular demographic such as gender, age, geographic location, income, occupation, etc.

Typically, a monitoring entity is an audience measurement entity (AME) that did not provide the media to the client devices and that is a trusted (e.g., neutral) third party for providing accurate usage statistics (e.g., The Nielsen Company, LLC). Since the AME is a third party, a cookie sent to the AME in an impression request to report the occurrence of the media impression at the client device is a third-party cookie.

Third-party cookie tracking is used by measurement entities to track access to media accessed by client devices from first-party media servers. For example, third-party cookies can be used by third parties to estimate a unique audience size of media served to client devices by multiple web servers. As the use of third-party cookies becomes increasingly restricted by web browsers, the abilities of third parties to reliably monitor Internet-based media accesses using computing-based techniques that do not rely on human intervention becomes challenging. Humans cannot reliably analyze, filter, and confirm the vast quantity of collected media impression data for thousands or millions of media accesses to generate audience metrics information in a reasonable amount of time. To overcome such challenges and limitations of human intervention, examples disclosed herein provide computing-based techniques to overcome the restricted use of third-party cookies for audience metrics generation by deduplicating audience data between two database proprietors based on sketch data provided by each of the database proprietors. Such methods produce an estimate of the unique audience size while preserving privacies of subscribers of those database proprietors.

Techniques for monitoring user accesses to an Internet-accessible media, such as digital television (DTV) media and digital content ratings (DCR) media, have evolved significantly over the years. Internet-accessible media is also known as digital media. In the past, such monitoring was done primarily through server logs. In particular, entities serving media on the Internet would log the number of

requests received for their media at their servers. Basing Internet usage research on server logs is problematic for several reasons. For example, server logs can be tampered with either directly or via zombie programs, which repeatedly request media from the server to increase the server log counts. Also, media is sometimes retrieved once, cached locally and then repeatedly accessed from the local cache without involving the server. Server logs cannot track such repeat views of cached media. Thus, server logs are susceptible to both over-counting and under-counting errors.

The inventions disclosed in Blumenau, U.S. Pat. No. 6,108,637, which is hereby incorporated herein by reference in its entirety, fundamentally changed the way Internet monitoring is performed and overcame the limitations of the server-side log monitoring techniques described above. For example, Blumenau disclosed a technique wherein Internet media to be tracked is tagged with monitoring instructions. In particular, monitoring instructions are associated with the hypertext markup language (HTML) of the media to be tracked. When a client device requests the media, both the media and the monitoring instructions are downloaded to the client device. The monitoring instructions are, thus, executed whenever the media is accessed, be it from a server or from a cache. Upon execution, the monitoring instructions cause the client device to send or transmit monitoring information from the client device to a content provider site. The monitoring information is indicative of the manner in which content was displayed.

In some implementations, an impression request or ping request can be used to send or transmit monitoring information by a client device using a network communication in the form of a hypertext transfer protocol (HTTP) request. In this manner, the impression request or ping request reports the occurrence of a media impression at the client device. For example, the impression request or ping request includes information to report access to a particular item of media (e.g., an advertisement, a webpage, an image, video, audio, etc.). In some examples, the impression request or ping request can also include a cookie previously set in the browser of the client device that may be used to identify a user that accessed the media. That is, using first-party cookies set by database proprietors that serve media to client devices of subscribers, impression requests or ping requests cause monitoring data reflecting information about an access to the media to be sent from the client device that downloaded the media to one of the database proprietors in association with a first-party cookie set by the database proprietor to identify the client device and/or a user of the client device. Sending the monitoring data from the client device to the database proprietor is known as an impression request because the type of network communication employed is an HTTP request message that includes information (e.g., media identifier, advertisement campaign identifier, cookie identifier, etc.) about the media impression that occurred for the accessed media. The database proprietor can determine that the client accessed particular media in response to the database proprietor receiving the impression request associated with the particular media. Thus, based on the impression requests, the database proprietor can determine a subset of subscribers that accessed the particular media.

As used herein, an impression is defined to be an event in which a home or individual accesses and/or is exposed to media (e.g., an advertisement, content, a group of advertisements and/or a collection of content). In Internet media delivery, a quantity of impressions or impression count is the total number of times media (e.g., content, an advertisement,

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or advertisement campaign) has been accessed by a web population (e.g., the number of times the media is accessed). In some examples, an impression or media impression is logged by an impression collection entity (e.g., a database proprietor) in response to an impression request from a user/client device that requested the media. For example, an impression request is a message or network communication (e.g., an HTTP request) sent by a client device to an impression collection server to report the occurrence of a media impression at the client device. In some examples, a media impression is not associated with demographics. In non-Internet media delivery, such as television (TV) media, a television or a device attached to the television (e.g., a set-top-box or other media monitoring device) may monitor media being output by the television. The monitoring generates a log of impressions associated with the media displayed on the television. The television and/or connected device may transmit impression logs to the impression collection entity to log the media impressions.

A user of a computing device (e.g., a mobile device, a tablet, a laptop, etc.) and/or television may be exposed to the same media via multiple database proprietors. For example, a user may receive an advertisement while on a webpage of one database proprietor and receive the same advertisement through a web page of another database proprietor. In such an example, the exposures to the advertisement are logged by both database proprietors, resulting in the same audience member being counted twice as part of an audience of the media across both database proprietors. Multiple logged impressions associated with the same user for the same media item are problematic in determining total reach estimates because a user's multiple accesses to the same media via two or more media sources (e.g., two or more database proprietors) may be counted as two or more unique audience members for that media item. As used herein, reach is a measure indicative of the demographic coverage achieved by media (e.g., demographic group(s) and/or demographic population(s) exposed to the media). For example, media reaching a broader demographic base will have a larger reach than media that reached a more limited demographic base. The reach metric may be measured by tracking impressions for known users (e.g., database proprietor subscribers) for which a database proprietor stores demographic information or can obtain demographic information. Deduplication is a process that is necessary to adjust cross-source media exposure totals.

As used herein, a unique audience is the number of unique audience members between two database proprietors. Additionally, as used herein, a unique audience is based on audience members distinguishable from one another. That is, a particular audience member exposed to particular media is measured as a single unique audience member regardless of how many times that audience member is exposed to that particular media. If that particular audience member is exposed multiple times to the same media, the multiple exposures for the particular audience member to the same media is counted as only a single unique audience member. In this manner, impression performance for particular media is not disproportionately represented when a small subset of one or more audience members is exposed to the same media an excessively large number of times while a larger number of audience members is exposed fewer times or not at all to that same media. By tracking exposures to unique audience members, a unique audience measure may be used to determine a reach measure to identify how many unique audience members are reached by media. In some examples, increasing unique audience and, thus, reach, is useful for

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advertisers wishing to reach a larger audience base. In some examples disclosed herein, unique audience may also be referred to as unique audience size, audience size, deduplicated audience, and/or deduplicated audience size.

An AME may want to find unique audience sizes across multiple database proprietors. However, the protocols of the Internet make cookies inaccessible outside of the domain (e.g., Internet domain, domain name, etc.) on which they were set. Thus, a first-party cookie set in a media provider domain (e.g., the YouTube.com domain) is accessible to servers in that same domain, but not to servers outside that domain. Therefore, although an AME (e.g., a third party) might find it advantageous to access the first-party cookies set by the database proprietors, the AME is unable to do so. Although third-party cookies (e.g., cookies set by the AME) are useful for third-party measurement entities in many deduplication techniques to track media accesses and to leverage demographic information from third-party database proprietors, use of third-party cookies may be limited or may cease in some or all online markets. That is, use of third-party cookies enables sharing anonymous PII subscriber information across entities which can be used to identify and deduplicate audience members across database proprietor impression data. However, to reduce or eliminate the possibility of revealing user identities outside database proprietors by such anonymous data sharing across entities, some websites, internet domains, and/or web browsers will stop supporting third-party cookies. This will make it more challenging for third-party measurement entities to track media accesses via first-party servers. That is, although first-party cookies will still be supported and useful for media providers to track accesses to media via their own first-party servers, neutral third parties interested in generating neutral, unbiased audience metrics data will not have access to the impression data collected by the first-party servers using first-party cookies.

Examples disclosed herein provide computer-implemented technological solutions for overcoming such challenges by estimating audience sizes between two database proprietors using deduplication based on multiple vectors of counts. In examples disclosed herein, rather than the AME obtaining subscriber information via third-party cookies, the database proprietors are partnered with the AME to send to the AME sketch data that generally represents audience members that accessed media without revealing PII that could be used to identify true identities (e.g., names or aliases) of those audience members. In examples disclosed herein, the subscriber information includes a subset of subscribers from each database proprietor. In examples disclosed herein, each subset of subscribers includes those subscribers from the corresponding database proprietor who have accessed a particular media. Additionally or alternatively, the database proprietors can filter the subscribers represented in the sketch data based on audience demographics (e.g., gender, age, geographic location, income, occupation, etc.). In examples disclosed herein, a database proprietor preserves subscriber privacy by generating the sketch data as multiple vectors of counts before sending the sketch data to the AME for deduplication. In some examples, a first database proprietor sends its sketch data as multiple vectors of counts to a second database proprietor. In such examples, the second database proprietor uses the received sketch data and its own generated sketch data (e.g., multiple vectors of counts) to estimate audience size between the two database proprietors without involvement of the AME.

As used herein, a vector of counts represents an audience of a media item as a distribution of hash values corresponding to each subscriber (e.g., in a subset of subscribers represented by that vector of counts) of a database proprietor. In examples disclosed herein, the vector of counts includes a plurality of bins (e.g., elements). Each bin corresponds to a bin number. The hash value corresponding to a subscriber indicates the bin number for the subscriber. In examples disclosed herein, in response to a hash value indicating a given bin number, the value of the bin corresponding to the given bin number increments by one. As such, the value corresponding to a bin in the vector of counts indicates the number of subscribers assigned to that bin. Each of the database proprietors generates a plurality of vectors of counts to represent a selected audience of a media item under analysis. A vector of counts from a first database proprietor for a particular media item corresponds to a vector of counts from the second database proprietor for the same media item. Two corresponding vectors of counts between the database proprietors is a vector of counts pair. The first database proprietor generates a first vector of the vector of counts pair using a given number of bins and a given hashing algorithm. Similarly, the second database proprietor generates a second vector of the vector of counts pair using the same given number of bins and the same given hashing algorithm.

As used herein, a hash value is a number and/or character representation of an identifier of a subscriber (e.g., a subscriber identifier). In examples disclosed herein, the subscriber identifier uniquely identifies the subscriber to a database proprietor. Example information that may serve as a subscriber identifier includes an email address, a street address, a person's name, a telephone number, a username, a birth date, an Internet Protocol (IP) address, and/or any other information that can be used to uniquely identify a person and/or any combination thereof. In examples disclosed herein, a hashing algorithm converts the subscriber identifier to the corresponding hash value. Typically, the hash value is a binary value, a hexadecimal value, or a decimal value. The hash value transforms to a bin number in the vector of counts.

As used herein, a binary hash (also referred to herein as a bit hash) is a hash function that generates a vector or array of multiple binary outputs (e.g., a string of 0s and/or 1s) from any input (e.g., an audience member's email address) with each element in the array being equally likely (e.g., there is a 50% chance that any given bit in the array will be a 0 and a 50% chance that the bit will be a 1). In some examples, the hash may generate an output that is not in binary form but may be converted to binary form with just 0s and 1s. For example, the hash may generate the hexadecimal output of D447, which would convert to the binary array of 1101010001000111. Further, the hash is defined such that any combination of array values generated by the hash function (e.g., any combination of 0s and 1s or other digits and/or letters for non-binary outputs) is equally likely as any other combination of array values. Further still, the hash is defined such that the same output will always result from the same input. As such, if both database proprietors **106a-b** use the same hash function on the same PII (e.g., "johnsmith@email.com"), both database proprietors **106a-b** will output the same array of values.

In some examples, to generate the final vector of counts sketch data, the output of the hash function applied to each entry in the database is used to generate an integer value from 1 to n, where n is the number of elements in the vector of counts. The transformation of the output of the hash

function to the integer value may be accomplished in any manner that results in a distribution that is uniform (e.g., there is a 1/n probability for any given integer value resulting from the transformation) and consistent (e.g., the same output of the hash function always produces the same integer value). The integer value derived from the output of the hash function applied to a particular data entry is used to identify the particular element within the vector of counts that is to be incremented by 1 to represent the particular data entry. For example, if a first entry in a database (e.g., "johnsmith@email.com") is hashed to the binary array of 11010100, and the total number of elements in the vector of counts is n=16, then only four bits are needed to define any integer value from 1 to 16. Accordingly, in some examples, the four leading bits of 1101 are selected to define a base ten number corresponding to the integer value. That is, the binary value of 1101 corresponds to the decimal number 13 such that the 13th element (out of 16) in the vector of counts is incremented by 1. As subsequent entries in the database are hashed and transformed to an integer value, the corresponding element in the vector of counts is incremented such that each element in the final vector will represent a count of the total number of entries designated to each element in the vector. As a result, the summation of values across all elements in the vector of counts will correspond to the cardinality of the sketch data (e.g., the total number of unique entries in the database represented by the vector of counts). In some examples, multiple different hash functions may be applied to each entry and assigned to a particular element within the vector of counts based on the process outlined above. In such examples, the summation of values across all elements in the vector of counts will correspond to the cardinality of the sketch data multiplied by the number of hash functions used. For purposes of explanation, examples described below assume only one hash function is used.

Deduplicating based on multiple vectors of counts, as disclosed herein, involves determining a first variance for the first vectors of counts of the first database proprietor, determining a second variance for the second vectors of counts of the second database proprietor, and determining a covariance for the first vectors of counts and the second vectors of counts. In addition, examples disclosed herein determine an estimate of the deduplicated value by determining the solution to a polynomial based on the first variance, the second variance, and the covariance. The deduplicated value is indicative of the audience size for an item of media (e.g., the unique number of people that accessed a particular media item).

Although examples disclosed herein are described in association with audience metrics related to media impressions, examples disclosed herein may be similarly used for other applications to deduplicate between any two data sets while preserving privacy. The data sets themselves need not be audiences or email addresses. They could be, for example, bank accounts, lists of purchased items, store visits, traffic patterns, etc. The data sets could be represented as lists of numbers or any other information.

FIG. 1 shows an example operating environment **100** that includes an example audience measurement entity (AME) **102**, an example database proprietor A **106a**, an example database proprietor B **106b**, and example client devices **108**. The example AME **102** includes an example AME computer **110** that implements an example audience metrics generator **112** to determine audience sizes based on sketch data generated by the database proprietors **106a,b**. In some examples, the example audience metrics generator **112** can

be implemented by at least one of the database proprietors **106a,b** instead of the example AME **102**. In the illustrated example of FIG. 1, the client devices **108** may be stationary or portable computers, handheld computing devices, smart phones, Internet appliances, and/or any other type of device that may be connected to the Internet and capable of accessing and/or presenting media.

As used herein, an audience size is defined as a number of deduplicated or unique audience members exposed to a media item of interest for audience metrics analysis. A deduplicated or unique audience member is one that is counted only once as part of an audience size. Thus, regardless of whether a particular person is detected as accessing a media item once or multiple times, that person is only counted once in the audience size for that media item. Audience size may also be referred to as unique audience, unique audience size, deduplicated audience, and/or deduplicated audience size.

As used herein, a media impression is defined as an occurrence of access and/or exposure to media **114** (e.g., an advertisement, a movie, a movie trailer, a song, a web page banner, etc.). Examples disclosed herein may be used to monitor for media impressions of any one or more media types (e.g., video, audio, a web page, an image, text, etc.). In examples disclosed herein, the media **114** may be content and/or advertisements. Examples disclosed herein are not restricted for use with any particular type of media. On the contrary, examples disclosed herein may be implemented in connection with tracking impressions for media of any type or form in a network.

In the illustrated example of FIG. 1, content providers and/or advertisers distribute the media **114** via the Internet to users that access websites and/or online television services (e.g., web-based TV, Internet protocol TV (IPTV), etc.). In some examples, the media **114** is served by media servers of the same internet domains as the database proprietors **106a,b**. For example, the database proprietors **106a,b** include corresponding database proprietor servers **118a,b** that can serve media **114** to their corresponding subscribers via the client devices **108**. Examples disclosed herein can be used to generate audience metrics data that measures audience sizes of media served by different ones of the database proprietors **106a,b**. For example, the database proprietors **106a,b** may use such audience metrics data to promote their online media serving services (e.g., ad server services, media server services, web server services, etc.) to prospective clients. By showing audience metrics data indicative of audience sizes drawn by corresponding ones of the database proprietors **106a,b**, the database proprietors **106a,b** can sell their media serving services to customers interested in delivering online media to users.

The media **114** is then presented via the client devices **108**. When the media **114** is accessed by the client devices **108**, the client devices **108** send impression requests **122a,b** to the database proprietor servers **118a,b** to inform the database proprietor servers **118a,b** of the media accesses. In this manner, the database proprietor servers **118a,b** can log media impressions in impression records of corresponding database proprietor audience metrics databases **124a,b**. When a database proprietor server **118a,b** serves the media **114**, the impression request **122a,b** includes a first-party cookie set by that database proprietor server **118a,b** so that the database proprietor server **118a,b** can log an impression for the media **114** without using a third-party cookie. In some examples, the database proprietor servers **118a,b** may additionally or alternatively log impressions based on requests for media **114** from the client devices **108**. For

example, when a user of client device **108** provides a URL or selects an item of media for viewing, the client device **108** sends an HTTP request to a database proprietor server **118a,b** that includes the first-party cookie and an identifier of the requested media. In response, the database proprietor server **118a,b** serves the requested media to that client device **108** and logs an impression of the media as attributable to the client device **108**. In the illustrated example of FIG. 1, the database proprietors **106a,b** log demographic impressions corresponding to accesses by the client devices **108** to the media **114**. Demographic impressions are impressions logged in association with demographic information (e.g., PII) collected by the database proprietors **106a,b** from registered subscribers of their services. Examples disclosed herein are based on audience data from impressions collected by the database proprietors **106a,b**.

In some examples, the media **114** is encoded to include a media identifier (ID). The media ID may be any identifier or information that can be used to identify the corresponding media **114**. In some examples the media ID is an alphanumeric string or value. In some examples, the media ID is a collection of information. For example, if the media **114** is an episode, the media ID may include program name, season number, and/or episode number. When the example media **114** includes advertisements, such advertisements may be content and/or advertisements. The advertisements may be individual, standalone ads and/or may be part of one or more ad campaigns. The ads of the illustrated example are encoded with identification codes (e.g., data) that identify the associated ad campaign (e.g., a campaign ID, if any), a creative type ID (e.g., identifying a Flash-based ad, a banner ad, a rich type ad, etc.), a source ID (e.g., identifying the ad publisher), and/or a placement ID (e.g., identifying the physical placement of the ad on a screen). In some examples, advertisements tagged with the monitoring instructions are distributed with Internet-based media content such as, for example, web pages, streaming video, streaming audio, IPTV content, etc. As noted above, methods, apparatus, systems, and/or articles of manufacture disclosed herein are not limited to advertisement monitoring but can be adapted to any type of content monitoring (e.g., web pages, movies, television programs, etc.).

In some examples, the media **114** of the illustrated example is tagged or encoded to include monitoring or tag instructions, which are computer-executable monitoring instructions (e.g., Java, java script, or any other computer language or script) that are executed by web browsers that access the media **114** via, for example, the Internet. Execution of the monitoring instructions causes the web browser to send the impression requests **122a,b** (e.g., also referred to as tag requests) to the database proprietor A **106a**, and/or the database proprietor B **106b**. As used herein, tag requests **122a,b** are used by the client devices **108** to report occurrences of media impressions caused by the client devices accessing the media **114**. In the illustrated example, the tag requests **122a,b** include user-identifying information that the database proprietors **106a,b** can use to identify the subscriber that accessed the media **114**. For example, when a subscriber of the database proprietor **106a** logs into a server of the database proprietor **106a** via a client device **108**, the database proprietor **106a** sets a database proprietor cookie (e.g., a first-party cookie) on the client device **108** and maps that cookie to the subscriber's identity/account information at the database proprietor server **118a**. In examples disclosed herein, subscriber identity and/or subscriber account information includes personally identifiable information (PII) such as full name, street address, residence city and state,

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telephone numbers, email addresses, ages, dates of birth, social security numbers, demographic information, and/or any other personal information provided by subscribers in exchange for services from the database proprietors **106a,b**. By having such PII information mapped to database proprietor cookies, the database proprietor **106a** can subsequently identify the subscriber based on the database proprietor cookie to determine when that user accessed different media **114** and to log an impression in association with demographics and/or other PII information of that user. In the illustrated example of FIG. 1, the impression requests **122a-b** (e.g., tag requests) include database proprietor cookies of the client devices **108** to inform the database proprietors **106a,b** of the particular subscribers that accessed the media **114**.

The tag requests **122a,b** may be implemented using HTTP requests. However, whereas HTTP requests are network communications that traditionally identify web pages or other resources to be downloaded, the tag requests **122a,b** of the illustrated example are network communications that include audience measurement information (e.g., ad campaign identification, content identifier, and/or user identification information) as their payloads. That is, the example tag requests **122a,b** are dummy HTTP requests that do not return web pages or resources. The server (e.g., the database proprietor servers **118a,b**) to which the tag requests **122a,b** are directed is programmed to log occurrences of impressions reported by the tag requests **122a,b** in, for example, impression records. Further examples of monitoring instructions (e.g., beacon instructions) and uses thereof to collect impression data are disclosed in U.S. Pat. No. 8,370,489 entitled "Methods and Apparatus to Determine Impressions using Distributed Demographic Information," which is hereby incorporated herein by reference in its entirety.

In other examples in which the media **114** is accessed by apps on mobile devices, tablets, computers, etc. (e.g., that do not employ cookies and/or do not execute instructions in a web browser environment), an app publisher (e.g., an app store) can provide a data collector in an install package of an app for installation at the client devices **108**. When a client device **108** downloads the app and consents to the accompanying data collector being installed at the client device **108** for purposes of audience/media/data analytics, the data collector can detect when the media **114** is accessed at the client device **108** and causes the client device **108** to send one or more of the impression requests **122a,b** to report the access to the media **114**. In such examples, the data collector can obtain user identifiers and/or device identifiers stored in the client devices **108** and send them in the impression requests **122a,b** to enable the database proprietors **106a,b** to log impressions. Further examples of using a collector in client devices to collect impression data are disclosed in U.S. Pat. No. 8,930,701 entitled "Methods and Apparatus to Collect Distributed User Information for Media Impressions and Search Terms," and in U.S. Pat. No. 9,237,138 entitled "Methods and Apparatus to Collect Distributed User Information for Media Impressions and Search Terms," both of which are hereby incorporated herein by reference in their entireties.

In the illustrated example, the database proprietors **106a,b** would like to collaborate with the AME **102** so that the AME **102** can operate as an independent party that measures and/or verifies audience measurement information pertaining to the media **114** accessed by the subscribers of the database proprietors **106a,b**. However, the database proprietors **106a,b** desire to do so while protecting the privacies of their subscribers by not sharing or revealing subscriber identities, subscriber information, and/or any other sub-

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scriber PII information to outside parties. In examples disclosed herein, to share impression data with the AME **102** without revealing subscriber identities, subscriber information, and/or any other subscriber PII information, the database proprietors **106a,b** process their collected impression data to generate corresponding sketch data **132a-b** as described below in connection with FIG. 2.

As used herein, sketch data is an arrangement of data for use in massive data analyses. For example, operations and/or queries that are specified with respect to the explicit and/or very large subsets of data can be processed much faster from the smaller sketches representing the actual data than if processed based on the actual, larger data space. This enables processing each observed item of data (e.g., each logged media impression and/or audience member) quickly in order to create a summary of the current state of the actual data. In some examples, the time to process a query using sketch data can be orders of magnitude less than the time to process a query using the actual data.

FIG. 2 is an example sketch data generation process **200** implemented by the database proprietors **106a,b** of FIG. 1 to provide the sketch data **132a,b** (FIG. 1) to the AME computer **110** (FIG. 1) for use in estimating audience sizes of media (e.g., the media **114** of FIG. 1). In the illustrated example, the database proprietors **106a,b** are in communication with the AME computer **110** to cooperate with the AME **102** to determine unique audience sizes for different media items accessed via the Internet on client devices (e.g., the client devices **108** of FIG. 1) by user subscribers of the database proprietors **106a,b**. For example, the database proprietors **106a,b** log impressions reported by client devices as described above in connection with FIG. 1 and leverage this data so that the AME **102** can generate audience metrics data without revealing user identities or personally identifiable information (PII) of the subscribers of the database proprietors **106a,b**. To protect privacy of user data while leveraging a separate AME **102** to determine unique audience sizes of different media items, the database proprietors **106a,b** generate the sketch data **132a,b** (e.g., subscriber data) to provide to the AME computer **110** as described below in connection with FIG. 2.

In the example of FIG. 2, each of the database proprietors **106a,b** includes a corresponding example hasher **202a,b** to produce the sketch data **132a,b** for delivery to the AME **102**. In the example of FIG. 2, the database proprietors **106a,b** agree on particular media and audience demographics for which to generate audience metrics. The example database proprietors **106a,b** filter their respective audience data (e.g., represented in the logged impressions) to only include audience members having accessed the particular media and belonging to the particular demographics. The example hashers **202a,b** include deduplicators **218a,b** that deduplicate audience data to generate unique audience data relative to each database proprietor **106a,b**. The database proprietors **106a,b** include hash data generators **222a,b** that use an agreed upon method of hashing and summarizing the deduplicated, unique audience data. In examples disclosed herein, the agreed upon type of hashing and summary type is vector of counts sketch data. In examples disclosed herein, each of the database proprietors **106a,b** produces sketch data for multiple vectors of counts. In some examples, one of the database proprietors **106a,b** generates vectors of counts of different lengths relative to the vectors of counts generated by the other database proprietor **106a,b**. In the example of FIG. 2, the hash data generators **202a,b** perform the agreed upon hashing and summarization to generate corresponding raw unedited hashed data **226a,b**. In the illustrated example

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of FIG. 2, the database proprietors **106a,b** may also elect to insert noise into the raw unedited hashed data **226a,b** using noise inserters **228a,b** to produce data with a secondary level of privacy protection for subscribers' PII data. However, in some examples, the database proprietors **106a,b** may elect not to insert noise. The example hash data generators **202a,b** also include privacy generators **230a,b** to zero-mean the raw data **226a,b** (with noise or without noise) to generate the sketch data **132a,b**. In some examples, the database proprietors **106a,b** may elect not to zero-mean the raw data **226a,b**. In such examples, the sketch data **132a,b** is provided to the AME **102** as raw data.

FIG. 3 illustrates the example audience metrics generator **112** of FIGS. 1 and 2 to estimate unique audience sizes of media **114** (FIG. 1) using deduplication based on sketch data **132a,b** that includes multiple vectors of counts representative of the subscribers of the database proprietors **106a,b**. In the illustrated example, the audience metrics generator **112** receives the sketch data A **132a** from the database proprietor A **106a**, and receives the sketch data B **132b** from the database proprietor B **106b**. To access and process the sketch data **132a,b**, the example audience metrics generator **112** includes an example data interface **304**, an example coefficient generator **306**, an example variance detector **308**, an example real roots solver **310**, an example data analyzer **312**, an example noise detector **314**, an example numerical solver **315**, and an example audience size generator **316**.

In the illustrated example of FIG. 3, the audience metrics generator **112** is provided with the data interface **304** to access the sketch data **132a,b**. For example, the data interface **304** may be a network interface to access the sketch data **132a,b** via network communications from servers of the database proprietors **106a,b**. Alternatively, the example data interface **304** may be a memory/storage interface to access the sketch data **132a,b** from memory and/or data storage devices.

In examples disclosed herein, each database proprietor **106a,b** provides multiple vectors of counts (e.g., two or more vectors of counts). Also in examples disclosed herein, a pair of vectors of counts refers to a first vector of counts (e.g., $V_A^{(1)}$) from one database proprietor **106a,b** and a second vector of counts (e.g., $V_B^{(1)}$) from the other one of the database proprietors **106a,b**. Since multiple vectors of counts are provided by each database proprietor **106a,b**, there are multiple pairs of vectors of counts (e.g., $V_A^{(1)}$ and $V_B^{(1)}$; $V_A^{(2)}$ and $V_B^{(2)}$; etc.). In addition, bin count (e.g., number of bins in a vector of counts, vector of counts length) may be different between pairs of vectors of counts. For example, one pair may have vectors of counts of five bins in length (e.g., $V_A^{(1)}, V_B^{(1)} = \{\text{bin 1, bin 2, bin 3, bin 4, bin 5}\}$) and another pair may have vectors of counts of 10 bins in length (e.g., $V_A^{(2)}, V_B^{(2)} = \{\text{bin 1, bin 2, bin 3, bin 4, bin 5, bin 6, bin 7, bin 8, bin 9, bin 10}\}$). In examples disclosed herein, the database proprietors **106a,b** select the bin count for each pair of vectors of counts. A high bin count increases the accuracy of estimations of audience size, but reduces subscriber privacy. Alternatively, a low bin count reduces the accuracy of estimations of audience size, but increases subscriber privacy. As such, the database proprietors **106a,b** select the appropriate bin counts based on the size of the subscriber base of each database proprietor **106a,b**, a target accuracy for audience size estimation, and/or a target level or strength of privacy for their subscribers. In some examples, the database proprietors **106a,b** can split a single vector of counts (e.g., a vector of counts with a bin count of 1000) into multiple vectors of counts (e.g., 10 vectors of counts with bin counts of 100) to increase subscriber privacy.

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In the illustrated example of FIG. 3, the audience metrics generator **112** is provided with the data analyzer **312** to analyze the sketch data **132a,b** from the data interface **304**. The example data analyzer **312** first checks if the sketch data **132a,b** is raw data and/or zero-meaned data. In some examples, the database proprietor A **106a** may provide raw data and the database proprietor B **106b** may provide zero-meaned data. The audience metrics generator **112** can process the sketch data **132a,b** regardless of whether it is provided as raw data or as zero-meaned data. The example data analyzer **312** has multiple functions including, for example: (a) determining if each pair of vectors of counts in the incoming sketch data **132a,b** from the data interface **304** is raw data and/or zero-meaned data, (b) deciding if the data analyzer **312** should zero-mean the raw data in each pair of vectors of counts, (c) passing the sketch data **132a,b** to the example variance detector **308**, and (d) alerting the example variance detector **308** which set of equations to use, as discussed below. For example, if both sets of the sketch data **132a,b** are provided as raw data, the data analyzer **312** can zero-mean the sketch data **132a,b** before proceeding with the calculations.

In the illustrated example, depending on whether the sketch data **132a,b** is provided as raw data or zero-meaned data, the variance detector **308** can find the variance of the sketch data **132a,b** using zero-meaned data equations for each pair of vectors of counts. In the illustrated example, Equations 1 through 10 below are used for zero-meaned data. In the illustrated example, in response to the data analyzer **312** determining that the sketch data **132a,b** is raw data, the data analyzer **312** can zero mean the sketch data **132a,b** before the variance detector **308** determines the variance of the sketch data **132a,b** using the zero-meaned data equations (e.g., Equations 1 through 10).

In examples disclosed herein in which the sketch data **132a,b** includes multiple vectors of counts, each pair of vectors of counts includes a first vector of counts from the sketch data **132a**, and a second vector of counts from the sketch data **132b**. For each pair of vectors of counts, the example variance detector **308** generates a first variance based on Equation 1 below and a second variance based on Equation 2 below. The first variance corresponds to the first vector of counts from the sketch data A **132a**. The second variance corresponds to the second vector of counts from sketch data B **132b**. In such examples, the example variance detector **308** also determines a covariance for each pair of vector of counts based on Equation 3 below. Additionally, in some examples, the variance detector **308** weights each of the first and second variances and the covariance using a constant term related to the length of vectors (e.g., bin count, number of bins per vector) in a pair. The example variance detector **308** then generates weighted sum of variances and a weighted sum of covariances based on Equations 4, 5, and 6 below. Using the weighted sum of variances and the weighted sum of covariances, the example variance detector **308** may determine normalized weighted sum of variances and normalized weighted sum of covariances between sketch data A **132a** and sketch data B **132b** based on Equations 8, 9, and 10 below.

$$S_{xx}^{(j)} = \sum_{i=1}^{n^{(j)}} (x_i^{(j)})^2$$

Equation 1

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-continued

$$S_{yy}^{(j)} = \sum_{i=1}^{n^{(j)}} (y_i^{(j)})^2$$

Equation 2

$$S_{xy}^{(j)} = \sum_{i=1}^{n^{(j)}} (x_i^{(j)})(y_i^{(j)})$$

Equation 3

$$S_{xx}' = \sum_{j=1}^m \frac{S_{xx}^{(j)}}{c^{(j)}}$$

Equation 4

$$S_{yy}' = \sum_{j=1}^m \frac{S_{yy}^{(j)}}{c^{(j)}}$$

Equation 5

$$S_{xy}' = \sum_{j=1}^m \frac{S_{xy}^{(j)}}{c^{(j)}}$$

Equation 6

$$c^{(j)} = p^{(j)}(1 - p^{(j)})$$

Equation 7

$$V_{xx}' = \frac{S_{xx}'}{N} = \frac{1}{N} \sum_{j=1}^m \frac{S_{xx}^{(j)}}{c^{(j)}}$$

Equation 8

$$V_{yy}' = \frac{S_{yy}'}{N} = \frac{1}{N} \sum_{j=1}^m \frac{S_{yy}^{(j)}}{c^{(j)}}$$

Equation 9

$$V_{xy}' = \frac{S_{xy}'}{N} = \frac{1}{N} \sum_{j=1}^m \frac{S_{xy}^{(j)}}{c^{(j)}}$$

Equation 10

In the above Equations, m refers to the number of vectors of count pairs across both the sketch data A **132a** and the sketch data B **132b**, where j is the index of the vector pair (e.g., j={1, . . . , m}). In the above Equation 1, $x_i^{(j)}$ refers to the bin value for bin i in the first vector of counts in pair j corresponding to the sketch data A **132a**. In the above Equation 2, $y_i^{(j)}$ refers to the bin value for bin i in the second vector of counts in pair j corresponding to the sketch data B **132b**. $S_{xx}^{(j)}$ refers to the first variance of the bin values $x_i^{(j)}$ in the first vector of counts in pair j corresponding to the sketch data A **132a**. $S_{yy}^{(j)}$ refers to the second variance of the bin values $y_i^{(j)}$ in the second vector of counts in pair j corresponding to the sketch data B **132b**. In Equation 3, $S_{xy}^{(j)}$ refers to the covariance of the bin values $x_i^{(j)}$, $y_i^{(j)}$ between the first vector of counts and the second vector of counts in pair j. The term $n^{(j)}$ is the length of the vectors of counts in pair j (e.g., the number of bins in each of vector of counts in pair j). As $n^{(j)}$ increases, the accuracy of the estimate for the unique audience increases. In the above Equation 4, S_{xx}' refers to the first weighted sum of variances of the bin values $x_i^{(j)}$ for the sketch data A **132a**. In the above Equation 5, S_{yy}' refers to the second weighted sum of variances of the bin values $y_i^{(j)}$ for the sketch data B **132b**. In the above Equation 6, S_{xy}' refers to the weighted sum of covariances of the bin values $x_i^{(j)}$, $y_i^{(j)}$ for the sketch data **132a,b**. In the above Equations 4, 5 and 6, $c^{(j)}$ refers to a probability factor for vectors of counts pair j. The example probability factor $c^{(j)}$ (represented by lower-case c) is used to weight the weighted sums of variances and the weighted sum of covariances to normalize the contributions of vectors of counts of different lengths to the weighted sums of variances and the weighted sums of covariances. In Equation 7 above, the probability factor $c^{(j)}$ is based on a probability $p^{(j)}$ of a subscriber being assigned to a given bin in a vector of counts of pair j. In examples disclosed herein, the probability $p^{(j)}$ is based on the length of each vector of counts in pair j, where $p^{(j)}=1/n^{(j)}$. In the above Equation 8, V_{xx}' refers to the normalized weighted sum of variances of the sketch data A **132a**. In the above Equation 9, V_{yy}' refers

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to the normalized weighted sum of variances of the sketch data B **132b**. In the above Equation 10, V_{xy}' refers to the normalized weighted sum of covariances of the sketch data A **132a** and the sketch data B **132b**. In Equations 8, 9, and 10 above, N refers to the sum of the lengths of all vectors of counts in the sketch data A **132a** or the sketch data B **132b**. |A| and |B| refer to the cardinality of the sketch data A **132a** and the sketch data B **132b**. In examples disclosed herein, the variance is a measure of variability in a list of numbers (e.g., bin values in the vectors of counts of the sketch data A **132a** and/or the vectors of counts of the sketch data B **132b**). In examples disclosed herein, the covariance is a measure of variability in two lists of numbers (e.g., bin values between bins for vectors of counts of the sketch data A **132a** and the sketch data B **132b**). In examples disclosed herein, the cardinality is the number of unique elements (e.g., subscribers) in the sketch data A **132a**, in the sketch data B **132b**, and/or between the sketch data A **132a** and the sketch data B **132b**.

Equations 1 through 10 above are used to determine the variances, the covariance, the weighted sum of variances, the weighted sum of covariances, the normalized weighted sum of variances, and the normalized weighted sum of covariances for zero-measured data. The above equations can also be used for data that is not zero-measured (e.g., raw data). In response to the variance detector **308** determining the normalized weighted sums of variances and the normalized weighted sum of covariance, the example variance detector **308** provides the normalized weighted sums of variances and normalized weighted sums of covariances and cardinality to the example noise checker **314**.

The example noise checker **314** accounts for any noise that may have been intentionally inserted by the database proprietors **106a,b** in their hashing of the raw data **226a,b** (FIG. 2). Each database proprietor **106a,b** may add its own noise independent of the other database proprietor **106a,b**. For example, different degrees/types of noise may be added by the database proprietors **106a,b** and/or one database proprietor **106a,b** inserts noise while the other does not. When each database proprietor **106a,b** provides multiple vectors of counts, the degree/types of noise added by each database proprietor **106a,b** may also differ between the multiple vectors of counts. In the illustrated example, the database proprietors **106a,b** inform the AME **102** of the value of the variance for the inserted noise distribution for each vector of counts. The example noise checker **314** sets noise parameters $V_1^{(j)}$ and $V_2^{(j)}$ of Equation 17 below to the respective inserted noise variance values from the database proprietors **106a,b**. That is, each database proprietor **106a,b** can choose its own noise distribution, no matter how complex, with its own set of parameters values, and can provide the value of the variance for each vector of counts to the AME **102** without needing to provide all the parameters of the noise distribution. In this manner, the AME **102** can use the variance of the noise for each vector of counts in each pair j ($V_1^{(j)}$ or $V_2^{(j)}$) in the estimation formulas for W (e.g., where W is the unique elements contained in both sketch data A **132a** and sketch data B **132b**). For example, this parameter may be the location and scale parameters for a Laplacian, the mean and variance of a Gaussian distribution, and/or even a complicated 5-parameter distribution. In the illustrated example, the database proprietors **106a,b** can choose any noise distribution. When the parameters are numerically defined by the database proprietors **106a,b**, a scalar singular number results. The scalar singular number is the variance for the distribution with the numerical values defined for the parameters. The scalar singular number is

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provided to the example noise checker **314**. When no noise is present, the example noise checker **314** inputs a value of zero for the noise parameters $V_1^{(j)}$ and $V_2^{(j)}$ (e.g., if no noise parameter is given by the database proprietor **106a,b** and an assumption of noise is not made). These noise terms are added to further obfuscate the sketch data **132a,b** at a user level. This means that the noise terms make it harder for a third-party entity (e.g., the AME **102**) to find out the identity of each specific element, which in turn protects privacies of PII data of subscribers of the database proprietors **106a,b**. The example noise checker **314** accounts for the noise by selecting an equation (e.g., Equation 17 below) that accounts for noise for use by the coefficient generator **306** in subsequent operations.

The example audience metrics generator **112** is provided with the coefficient generator **306** to use the normalized weighted sums of variances, the normalized weighted sum of covariances, and the cardinality of the sketch data **132a,b** to generate coefficients for a third-degree polynomial when there is no noise in the sketch data **132a,b**. In the illustrated example, the audience metrics generator **112** selects which equations to use for determining an estimation of audience size based on determinations by the noise checker **314** of whether the sketch data **132a,b** includes noise. For example, the audience metrics generator **112** selects Equations 11 through 15 below to determine an estimation of audience size in response to determining that the sketch data **132a,b** does not include noise. Alternatively, in response to determining that the sketch data **132a,b** includes noise, the audience metrics generator **112** selects Equation 17 to determine the estimation of audience size.

If the database proprietors **106a,b** notify the AME **102** that there is no noise in the sketch data **132a,b**, or the example noise checker **314** assumes that there is no noise in the sketch data **132a,b**, the noise checker **314** selects Equations 11, 12, 13, and 14 shown below for use by the coefficient generator **306** to determine values of coefficient terms.

$$C_0 = -|A||B|V_{xy}' \quad \text{Equation 11:}$$

In Equation 11 above, the cardinality $|A|$ of sketch data A **132a** is multiplied by the cardinality $|B|$ of sketch data B **132b**, and the resulting product is multiplied by the normalized weighted sums of covariances V_{xy}' found using Equation 10 above. This multiplication produces the coefficient term C_0 (represented by upper-case C). Under the no-noise condition, Equation 12 below is used to determine the value of the coefficient term C_1 .

$$C_1 = |A|V_{yy}' + |B|V_{xx}' - |A||B| \quad \text{Equation 12:}$$

In Equation 12 above, the cardinality $|A|$ is multiplied by the normalized weighted sum of variance of sketch data B **132b**, V_{yy}' , found using Equation 9 above. This product is then added to the product of the cardinality $|B|$ and the normalized weighted sum of variance of sketch data A **132a**, V_{xx}' , found using Equation 8. The product of the cardinality $|A|$ and the cardinality $|B|$ is subtracted from the sum of products (e.g., $|A|V_{yy}' + |B|V_{xx}' - |A||B|$). The result is the coefficient term C_1 . Under the no-noise condition, Equation 13 below is used to determine the value of the coefficient term C_2 .

$$C_2 = -V_{xy}' \quad \text{Equation 13:}$$

In Equation 13 above, the coefficient term C_2 is simply the normalized weighted sum of covariance of the sketch data set A **132a** and the sketch data set B **132b** multiplied by

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negative one. Under the no-noise condition, Equation 14 below is used to determine the value of the coefficient term C_3 .

$$C_3 = 1 \quad \text{Equation 14:}$$

In Equation 14 above, the coefficient term C_3 is simply equal to one.

The real roots solver **310** then accesses the coefficients and creates a polynomial which it can solve in accordance with Equation 15 as shown below.

$$C_0 + C_1W + C_2W^2 + C_3W^3 = 0 \quad \text{Equation 15:}$$

In Equation 15 above, the coefficients are each multiplied by W which represents a number of unique elements that overlap or are duplicated between the sketch data A **132a** and the sketch data B **132b** (e.g., overlapping unique audience members) contained in both. In this example, the real roots solver **310** solves the equation (e.g., using a commercial solver) and selects the real root value for W . For example, there might be multiplicity or complex roots, but the real root is selected as the answer that makes logical sense (e.g., indicates a positive real number for the number of overlapping unique audience members) as described in Equation 16 below.

$$UA = |A| + |B| - W \quad \text{Equation 16:}$$

In Equation 16 above, $|A|$ is the cardinality or unique audience in set A, and $|B|$ is the cardinality or unique audience in set B. Equation 16 above means the unique elements of A (e.g., $|A|$) with the unique elements of B (e.g., $|B|$), which may have some overlapping elements, has the elements that overlap or are in common in both sets (e.g., W), subtracted once which has the final result of unique audience (e.g., UA). For example, a complex number would not satisfy this requirement.

If the database proprietors **106a,b** notify the AME **102** of the presence of noise, or the example noise checker **314** assumes the presence of noise, there is no analogous expression for the polynomial coefficients found using Equations 11, 12, 13, and 14. For multiple vectors of counts, the degree of the polynomial increases and, as such, there is no simple expression for the polynomial. Instead, the noise checker **314** informs the coefficient generator **306** that no coefficient terms are to be generated. The example noise checker **314** informs the numerical solver **315** to obtain an estimate of W using Equation 17 below.

$$\text{Equation 17}$$

$$\frac{\partial}{\partial W} \ell(W | D) = \frac{\partial}{\partial W} \ell(W | \{x, y\}_{j=1, \dots, n}^{j=1, \dots, m}) = \sum_{j=1}^m - \frac{c^{(j)2} W (S_{yy}^{(j)} (|A|c^{(j)} + V_1^{(j)}) + S_{xx}^{(j)} (|B|c^{(j)} + V_2^{(j)}) - 2c^{(j)} S_{xy}^{(j)} W)}{((|A|c^{(j)} + V_1^{(j)}) (|B|c^{(j)} + V_2^{(j)}) - c^{(j)2} W^2)^2} \dots + \frac{c^{(j)2} n^{(j)} W + c^{(j)2} S_{yy}^{(j)}}{(|A|c^{(j)} + V_1^{(j)}) (|B|c^{(j)} + V_2^{(j)}) - c^{(j)2} W^2}$$

The right-hand side of Equation 17 above expresses the derivative of the sum of the log-likelihoods. In Equation 17 above, m refers to the number of pairs of vectors of counts across both the sketch data A **132a** and the sketch data B **132b**, where j is the index of the vector pair. $|A|$ is the cardinality or unique audience in set A, and $|B|$ is the cardinality or unique audience in set B. $V_1^{(j)}$ and $V_2^{(j)}$ are the variance of the noise for each vector of counts in each pair

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j corresponding to sketch data A **132a** and sketch data B **132b**, respectively. $S_{xx}^{(j)}$ and $S_{yy}^{(j)}$ refer to the first variance and the second variance for vectors of counts pair j, respectively, and $S_{xy}^{(j)}$ refers to the covariance for vectors of counts pair j. $n^{(j)}$ is the length of the vectors of counts in pair j (which is representative of the number of bins used in the vectors of counts in pair j). $c^{(j)}$ refers to the probability factor corresponding to vectors of counts pair j. When the derivative of the sum of the log-likelihoods for the vectors of counts is equal to zero (e.g.,

$$\frac{\partial}{\partial W} \ell(W | D) = 0).$$

Equation 17 above can be used to numerically solve for a value of W which represents an estimate of the overlap of audience members in both sets of sketch data **132a, b**. The numerical solver **315** obtains an estimate of the overlap W (where W is the overlap of audience members in both sets of sketch data **132a, b**). The example audience size generator **316** then accesses the unique audience estimate **318** and the overlap W and stores the unique audience estimate **318** and the overlap W in memory. The example audience size generator **316** can transform the resultant unique audience estimate data **318** into a form that a receiving party (e.g., a customer of audience metrics analytics services) can understand that may have both the overlap W and the unique audience UA listed which explains the demographics reached by the media. The audience size generator **316** can also solve for a variance of the estimate of the overlap W using Equation 18 below.

$$\text{Var}[W] = \frac{(AB - W^2)^2}{N(AB + W^2)} \quad \text{Equation 18}$$

In Equation 18 above, $\text{Var}[W]$ refers to the variance of the estimate of the overlap W, A refers to the cardinality of the sketch data A **132a**, B refers to the cardinality of the sketch data B **132b**, and N refers to the sum of the lengths of all vectors of counts in the sketch data A **132a** or the sketch data B **132b**.

A numerical example employing the above techniques involves the database proprietor A **106a** having 1000 unique subscribers (e.g., cardinality |A| equal to 1000), and the database proprietor B **106b** having 2000 unique subscribers (e.g., cardinality |B| equal to 2000). The subscribers are audience members that accessed particular media. The example database proprietor A **106a** provides the sketch data A **132a** (e.g., data set A) to the AME computer **110**, and the example database proprietor B **106b** provides the sketch data B **132b** (e.g., data set B) to the AME computer **110**. In this example, of each the data set A and the data set B includes two vectors of counts. The first vector of counts has a length of 5 and the second vector of counts has a length of 10. In this example, the first vector of counts for the data set A is $V_A^1 = \{191, 190, 212, 190, 217\}$ and the second vector of counts for the data set A is $V_A^2 = \{102, 109, 97, 107, 90, 121, 89, 83, 98, 104\}$. Further, the first vector of counts for the data set B is $V_B^1 = \{361, 379, 427, 431, 402\}$ and the second vector of counts for the data set B is $V_B^2 = \{206, 216, 206, 209, 166, 205, 166, 203, 197, 226\}$. In this example, the example data analyzer **312** determines that the vectors of counts are zero-measured data.

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In this example, using Equation 1 above, the example variance detector **308** generates a first variance for the first vector of counts V_A^1 of the data set A (e.g., $S_{xx}^{(1)} = 714$), and a first variance for the second vector of counts V_A^2 of the data set A (e.g., $S_{xx}^{(2)} = 1,114$). Further, using Equation 2 above, the example variance detector **308** generates a second variance for the first vector of counts V_B^1 of the data set B (e.g., $S_{yy}^{(1)} = 3,656$), and a second variance for the second vector of counts V_B^2 of the data set B (e.g., $S_{yy}^{(2)} = 3,440$). In this example, using Equation 3 above, the example variance detector **308** generates a covariance for the first vectors of counts V_A^1 and V_B^1 (e.g., $S = 609$), and a covariance for the second vectors of counts V_A^2 and V_B^2 (e.g., $S_{xy}^{(2)} = 1,079$).

In this example, using Equation 4 above, the example variance detector **308** generates a first weighted sum of variances ($S_{xx}' = 16,840.3$) for the data set A based on the first variances $S_{xx}^{(1)}$ and $S_{xx}^{(2)}$. Further, using Equation 5 above, the example variance detector **308** generates a second weighted sum of variances ($S_{yy}' = 61,072.2$) for the data set B based on the second variances $S_{yy}^{(1)}$ and $S_{yy}^{(2)}$. In this example, using Equation 6 above, the example variance detector **308** generates a weighted sum of covariances ($S_{xy}' = 15,795.1$) for the data set A and the data set B based on the covariances $S_{xy}^{(1)}$ and $S_{xy}^{(2)}$.

In this example, using Equation 8 above, the example variance detector **308** generates a first normalized weighted sum of variances ($V_{xx}' = 1,122.69$) for the data set A based on the first weighted sum of variances S_{xx}' . Further, using Equation 9 above, the example variance detector **308** generates a second normalized weighted sum of variances ($V_{yy}' = 4,071.48$) for the data set B based on the second weighted sum of variances S_{yy}' . In this example, using Equation 10 above, the example variance detector **308** generates a normalized weighted sum of covariances ($V_{xy}' = 1,053.01$) for the data set A and the data set B based on the weighted sum of covariances S_{xy}' .

In this example, the example noise checker **314** determines that no noise is present in the data set A and/or the data set B. As such, the example noise checker **314** sets noise parameters for the data set A ($V_1^{(1)}$ and $V_1^{(2)}$) and noise parameters for the data set B ($V_2^{(1)}$ and $V_2^{(2)}$) equal to zero. In this example, the example coefficient generator **306** determines values for coefficients ($c_0 = -2.10602 \times 10^9$, $c_1 = 4.31685 \times 10^6$, $c_2 = -1,053.01$, $c_3 = 1$) based on Equations 11 through 14 above. Further, the example real roots solver **310** solves for the roots of a third-degree polynomial using Equation 15 above based on the determined coefficients. In this example, the example real roots solver **310** selects the real root value of W as the estimate of the overlap ($W = 521.8$) of unique audience members between the data set A and the data set B. Further, using Equation 16 above, the example audience size generator **316** determines the unique audience estimate **318** based on the overlap W, the cardinality of data set A, and the cardinality of data set B.

In this example, the actual overlap W of unique audience members between the data set A and the data set B is 500. An estimate of the overlap using only the first pair of vectors of counts of length 5 is $W = 361.6$. An estimate of the overlap using only the second pair of vectors of counts of length 10 is $W = 608.8$. Both of these estimates for the overlap W are less accurate than the estimated overlap $W = 521.8$ determined using both pairs of vectors of counts. As such, using multiple pairs of vectors of counts can provide a more accurate unique audience estimate compared to using only a single pair of vectors of counts.

The estimate of the overlap using multiple pairs of vectors of counts or a single pair of vector of counts is unbiased.

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That is, for estimates of the overlap obtained across many trials, using one or more different hash functions and/or one or more different vector of counts sizes for each trial, an average of the estimates across all trials is close to the true overlap. Further, increasing the number of trials moves the average of the estimates across all trials closer to the true overlap. In this example, the audience size generator **316** can also determine a true variance of the estimate based on Equation 18 using the true value of the overlap (e.g., $W=500$). The true variance of the estimate using only the first pair of vectors of counts is $W=272,222.2$. The true variance of the estimate using only the second pair of vectors of counts is $W=136,111.1$. The true variance of the estimate using both pairs of vectors of counts is $W=90,740.7$. A variance of the estimate using multiple vectors of counts is less than a variance of the estimate using a single vector of counts (e.g., using only the first pair of vectors of counts or the second pair of vectors of counts). As such, a confidence in the estimate of the overlap using multiple pairs of vectors of counts is greater than a confidence in the estimate of the overlap using only a single pair of vectors of counts. In other words, an error between the estimate of the overlap and the true overlap is decreased when using multiple pairs of vectors of counts compared to a single pair of vectors of counts.

Examples disclosed herein use Maximum Likelihood. An example theorem of Maximum Likelihood is that an efficient estimator estimates the quantity of interest in some "best possible" manner. For example, by the Cramer-Rao bound, the maximum likelihood estimate is the lowest variance any unbiased estimate can be. Other unbiased estimators will have higher variance. Examples disclosed herein can use Fisher's Information to derive the variance of the unique audience estimate **318** (e.g., which is used in Cramer-Rao bound).

In examples disclosed herein, the sufficient statistic of a bivariate normal is the mean and covariance matrix. Given those two values, the entire distribution is determined. Using mean-centering (either given, or in the formula), the sufficient statistic is just the covariance matrix. Of the four terms within the 2×2 matrix, only three are needed (as one term appears twice). These are the normalized weighted sum of variances of the sketch data A **132a** (V_{xx}'), the normalized weighted sum of variances of the sketch data B **132b** (V_{yy}'), and the normalized weighted sum of covariances of both the sketch data A **132a** and the sketch data B **132b** (V_{xy}') (e.g., $\{V_{xx}', V_{yy}', V_{xy}'\}$).

Examples disclosed herein can use the complete sufficient statistics $\{V_{xx}', V_{yy}', V_{xy}'\}$ to the maximum information allowed (e.g., Fisher's information) to produce the smallest variance of the estimate possible (e.g., via Cramer-Rao bound). In some examples, it is the theoretically optimal estimate for the sketch data **132a,b** and model provided.

In examples disclosed herein, the coefficient generator **306** may implement means for determining coefficient values, the real roots solver **310** may implement means for determining a real root value, the audience size generator **316** may implement means for determining an audience size, the variance detector **308** may implement means for determining normalized weighted sums of variances and normalized weighted sums of covariances, the data analyzer **312** may implement means for zero-meaning data, the noise checker **314** may implement means for selecting equations, and the numerical solver **315** may implement means for determining an overlap of audience members.

While an example manner of implementing the example audience metrics generator **112** of FIG. 1 is illustrated in

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FIG. 3, one or more of the elements, processes and/or devices illustrated in FIG. 3 may be combined, divided, re-arranged, omitted, eliminated and/or implemented in any other way. Further, the example data interface **304**, the example coefficient generator **306**, the example variance detector **308**, the example real roots solver **310**, the example data analyzer **312**, the example noise checker **314**, the example numerical solver **315**, and the example audience size generator **316** and/or, more generally, the example audience metrics generator **112** of FIG. 3 may be implemented by hardware, software, firmware and/or any combination of hardware, software and/or firmware. Thus, for example, any of the example data interface **304**, the example coefficient generator **306**, the example variance detector **308**, the example real roots solver **310**, the example data analyzer **312**, the example noise checker **314**, the example numerical solver **315**, and the example audience size generator **316** and/or, more generally, the example audience metrics generator **112** could be implemented by one or more analog or digital circuit(s), logic circuits, programmable processor(s), programmable controller(s), graphics processing unit(s) (GPU(s)), digital signal processor(s) (DSP(s)), application specific integrated circuit(s) (ASIC(s)), programmable logic device(s) (PLD(s)) and/or field programmable logic device(s) (FPLD(s)). When reading any of the apparatus or system claims of this patent to cover a purely software and/or firmware implementation, at least one of the example data interface **304**, the example coefficient generator **306**, the example variance detector **308**, the example real roots solver **310**, the example data analyzer **312**, the example noise checker **314**, the example numerical solver **315**, and/or the example audience size generator **316** is/are hereby expressly defined to include a non-transitory computer readable storage device or storage disk such as a memory, a digital versatile disk (DVD), a compact disk (CD), a Blu-ray disk, etc. including the software and/or firmware. Further still, the example audience metrics generator **112** of FIG. 3 may include one or more elements, processes and/or devices in addition to, or instead of, those illustrated in FIG. 3, and/or may include more than one of any or all of the illustrated elements, processes and devices. As used herein, the phrase "in communication," including variations thereof, encompasses direct communication and/or indirect communication through one or more intermediary components, and does not require direct physical (e.g., wired) communication and/or constant communication, but rather additionally includes selective communication at periodic intervals, scheduled intervals, aperiodic intervals, and/or one-time events.

A flowchart representative of example hardware logic, machine readable instructions, hardware implemented state machines, and/or any combination thereof for implementing the audience metrics generator **112** of FIGS. 1-3 is shown in FIG. 4. The machine-readable instructions may be one or more executable programs or portion(s) of an executable program for execution by a computer processor such as the processor **612** shown in the example processor platform **600** discussed below in connection with FIG. 6. The program may be embodied in software stored on a non-transitory computer readable storage medium such as a CD-ROM, a floppy disk, a hard drive, a DVD, a Blu-ray disk, or a memory associated with the processor **612**, but the entire program and/or parts thereof could alternatively be executed by a device other than the processor **612** and/or embodied in firmware or dedicated hardware. Further, although the example program is described with reference to the flowchart illustrated in FIG. 4, many other methods of imple-

menting the example audience metrics generator **112** may alternatively be used. For example, the order of execution of the blocks may be changed, and/or some of the blocks described may be changed, eliminated, or combined. Additionally or alternatively, any or all of the blocks may be implemented by one or more hardware circuits (e.g., discrete and/or integrated analog and/or digital circuitry, an FPGA, an ASIC, a comparator, an operational-amplifier (op-amp), a logic circuit, etc.) structured to perform the corresponding operation without executing software or firmware.

The machine-readable instructions described herein may be stored in one or more of a compressed format, an encrypted format, a fragmented format, a compiled format, an executable format, a packaged format, etc. Machine-readable instructions as described herein may be stored as data (e.g., portions of instructions, code, representations of code, etc.) that may be utilized to create, manufacture, and/or produce machine executable instructions. For example, the machine-readable instructions may be fragmented and stored on one or more storage devices and/or computing devices (e.g., servers). The machine-readable instructions may require one or more of installation, modification, adaptation, updating, combining, supplementing, configuring, decryption, decompression, unpacking, distribution, reassignment, compilation, etc. in order to make them directly readable, interpretable, and/or executable by a computing device and/or other machine. For example, the machine readable instructions may be stored in multiple parts, which are individually compressed, encrypted, and stored on separate computing devices, wherein the parts when decrypted, decompressed, and combined form a set of executable instructions that implement a program such as that described herein.

In another example, the machine-readable instructions may be stored in a state in which they may be read by a computer, but require addition of a library (e.g., a dynamic link library (DLL)), a software development kit (SDK), an application programming interface (API), etc. in order to execute the instructions on a particular computing device or other device. In another example, the machine-readable instructions may need to be configured (e.g., settings stored, data input, network addresses recorded, etc.) before the machine-readable instructions and/or the corresponding program(s) can be executed in whole or in part. Thus, the disclosed machine-readable instructions and/or corresponding program(s) are intended to encompass such machine-readable instructions and/or program(s) regardless of the particular format or state of the machine-readable instructions and/or program(s) when stored or otherwise at rest or in transit.

The machine-readable instructions described herein can be represented by any past, present, or future instruction language, scripting language, programming language, etc. For example, the machine-readable instructions may be represented using any of the following languages: C, C++, Java, C#, Perl, Python, JavaScript, HyperText Markup Language (HTML), Structured Query Language (SQL), Swift, etc.

As mentioned above, the example process of FIG. 5 may be implemented using executable instructions (e.g., computer and/or machine readable instructions) stored on a non-transitory computer and/or machine readable medium such as a hard disk drive, a flash memory, a read-only memory, a compact disk, a digital versatile disk, a cache, a random-access memory and/or any other storage device or storage disk in which information is stored for any duration (e.g., for extended time periods, permanently, for brief

instances, for temporarily buffering, and/or for caching of the information). As used herein, the term non-transitory computer readable medium is expressly defined to include any type of computer readable storage device and/or storage disk and to exclude propagating signals and to exclude transmission media.

“Including” and “comprising” (and all forms and tenses thereof) are used herein to be open ended terms. Thus, whenever a claim employs any form of “include” or “comprise” (e.g., comprises, includes, comprising, including, having, etc.) as a preamble or within a claim recitation of any kind, it is to be understood that additional elements, terms, etc. may be present without falling outside the scope of the corresponding claim or recitation. As used herein, when the phrase “at least” is used as the transition term in, for example, a preamble of a claim, it is open-ended in the same manner as the term “comprising” and “including” are open ended. The term “and/or” when used, for example, in a form such as A, B, and/or C refers to any combination or subset of A, B, C such as (1) A alone, (2) B alone, (3) C alone, (4) A with B, (5) A with C, (6) B with C, and (7) A with B and with C. As used herein in the context of describing structures, components, items, objects and/or things, the phrase “at least one of A and B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, and (3) at least one A and at least one B. Similarly, as used herein in the context of describing structures, components, items, objects and/or things, the phrase “at least one of A or B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, and (3) at least one A and at least one B. As used herein in the context of describing the performance or execution of processes, instructions, actions, activities and/or steps, the phrase “at least one of A and B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, and (3) at least one A and at least one B. Similarly, as used herein in the context of describing the performance or execution of processes, instructions, actions, activities and/or steps, the phrase “at least one of A or B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, and (3) at least one A and at least one B.

As used herein, singular references (e.g., “a”, “an”, “first”, “second”, etc.) do not exclude a plurality. The term “a” or “an” entity, as used herein, refers to one or more of that entity. The terms “a” (or “an”), “one or more”, and “at least one” can be used interchangeably herein. Furthermore, although individually listed, a plurality of means, elements or method actions may be implemented by, e.g., a single unit or processor. Additionally, although individual features may be included in different examples or claims, these may possibly be combined, and the inclusion in different examples or claims does not imply that a combination of features is not feasible and/or advantageous.

FIG. 4 is a flowchart representative of machine-readable instructions which may be executed to implement the example audience metrics generator **112** of FIGS. 1-3 to estimate unique audience sizes of media using deduplication based on variance, covariance, and cardinality of multiple vectors of counts from the database proprietors **106a,b** of FIGS. 1-3. In the example of FIG. 4, the same media item (e.g., an advertisement, a movie, a movie trailer, a song, a web page banner, etc.) is presented by both of the database proprietors **106a,b**. The example AME computer **110** executes the instructions represented by the flowchart of FIG. 4 to generate the unique audience estimate **318** (FIG. 3) based on the sketch data A **132a** from the database

proprietor A 106a and the sketch data B 132b from the database proprietor B 106b. In this manner, the AME computer 110 can use the unique audience estimate 318 as a representation of the unique audience size reached by both of the database proprietors 106a,b serving the same media.

The program of FIG. 4 begins at block 410 at which the example data interface 304 (FIG. 3) accesses multiple vectors of counts in the sketch data A 132a and multiple vectors of counts in the sketch data B 132b. For example, the multiple vectors of counts in the sketch data A 132a can include $V_A^1 = \{\text{bin 1, bin 2, bin 3, bin 4, bin 5}\}$ and $V_A^2 = \{\text{bin 1, bin 2, bin 3, bin 4, bin 5, bin 6, bin 7, bin 8, bin 9, bin 10}\}$, and the multiple vectors of counts in the sketch data B 132b can include $V_B^1 = \{\text{bin 1, bin 2, bin 3, bin 4, bin 5}\}$ and $V_B^2 = \{\text{bin 1, bin 2, bin 3, bin 4, bin 5, bin 6, bin 7, bin 8, bin 9, bin 10}\}$. At block 412, the data analyzer 312 (FIG. 3) determines whether the vectors of counts in the sketch data 132a,b are raw data or zero-means data. In the illustrated example, if the sketch data 132a,b is raw data, the data analyzer 312 zero-means the sketch data 132a,b. If at block 412, the data analyzer 312 determines that the sketch data 132a,b is zero-means, control proceeds with the zero-means data (block 416).

At block 418, the example variance detector 308 (FIG. 3) determines the normalized weighted sum of variance of the vectors of counts in the sketch data A 132a (V_{xx}') and the normalized weighted sum of variance of the vectors of counts in the sketch data B 132b (V_{yy}'). Example instructions that may be used to implement block 418 to determine the normalized weighted sum of variance of the vectors of counts in the sketch data A 132a (V_{xx}') and the normalized weighted sum of variance of the vectors of counts in the sketch data B 132b (V_{yy}') are described below in connection with FIG. 5. At block 420, the example variance detector 308 determines the normalized weighted sum of covariances of the vectors of counts of both the sketch data A 132a and the sketch data B 132b (V_{xy}').

At block 422, the example noise checker 314 (FIG. 3) determines whether noise is present in the sketch data 132a,b. If no noise is present (block 422: NO), the example noise checker 314 sets the noise parameters (e.g., $V_1^{(f)}$ and/or $V_2^{(f)}$) equal to zero (block 424). The example noise checker 314 selects no-noise equations (block 425). For example, the example noise checker 314 selects Equations 11 through 15 above to be used for solving for determining the overlap W of unique audience members between the sketch data A 132a and the sketch data B 132b. If noise is present (block 422: YES), the example noise checker 314 sets the noise parameters equal to a non-zero value (block 426). The example noise checker 314 selects a noise-based equation (block 427). For example, the example noise checker 314 selects Equation 17 above to be used for solving for determining the overlap W of unique audience members between the sketch data A 132a and the sketch data B 132b. For example, the audience metrics generator 112 may receive non-zero values for noise for each vector of counts in the respective sketch data 132a,b from corresponding ones of the database proprietors 106a,b. The example noise checker 314 sets the noise parameters $V_1^{(f)}$ and/or $V_2^{(f)}$ of Equation 17 above equal to the received non-zero values.

If noise is not present, the example coefficient generator 306 (FIG. 3) determines values for coefficients as described above in connection with FIG. 3. The example real roots solver 310 (FIG. 3) solves for roots of a third-degree polynomial based on the coefficients (block 430). For example, the example real roots solver 310 may solve for the roots using Equation 15 as described above. The example

real roots solver 310 selects the real root value of the polynomial (block 432). For example, the example real roots solver 310 selects the real root of the polynomial by selecting the answer that makes logical sense (e.g., indicates a positive real number for the number of overlapping unique audience members) as described in Equation 16 above. If noise is present, at block 429, the example numerical solver 315 (FIG. 3) numerically solves for the number of overlapping elements in both sketch data A 132a and sketch data B 132b. For example, the example numerical solver 315 numerically solves for the number of unique audience members that overlap or are in common in both sketch data 132a,b by solving for the overlap W in Equation 17 above.

The example audience size generator 316 (FIG. 3) determines the unique audience estimate 318 (FIG. 3) based on an improved estimate for unique elements (e.g., unique audience members) in both the sketch data A 132a and the sketch data B 132b, and based on overlapping elements W (e.g., overlapping audience members) in both the sketch data A 132a and the sketch data B 132b (block 434). For example, the unique audience estimate 318 represents the unique audience size reached by the media 114 served by both database proprietors 106a,b. The example audience size generator 316 stores the unique audience estimate 318 in a memory (block 436). The example audience size generator 316 sends the unique audience estimate 318 to a customer computer (block 438) through, for example, one or more network communications. The example process of FIG. 4 then ends.

FIG. 5 is a flowchart representative of example machine-readable instructions which may be executed to implement the example audience metrics generator 112 of FIG. 3 to determine the normalized weighted sum of variance of sketch data A 132a and the normalized weighted sum of variance of sketch data B 132b. The example instructions of FIG. 5 may be used to implement block 418 of FIG. 4.

The example program of FIG. 5 begins at block 505 at which the example variance detector 308 (FIG. 3) generates a first variance ($S_{xx}^{(f)}$). For example, the example variance finder generates the first variance of the bin values from the multiple vectors of counts in sketch data A 132a using Equation 1 above. At block 510, the example variance detector 308 generates a second variance ($S_{yy}^{(f)}$). For example, the example variance detector 308 generates the second variance of the bin values from the multiple vectors of counts in sketch data B 132b using Equation 2 above. At block 515, the example variance detector 308 generates a covariance ($S_{xy}^{(f)}$). For example, the example variance detector 308 generates the covariance of the bin values from the multiple vectors of counts from the sketch data A 132a and the multiple vectors of counts from the sketch data B 132b using Equation 3 above.

At block 520, the example variance detector 308 generates a first weighted sum of variances (S_{xx}'). For example, the example variance detector 308 generates the first weighted sum of variances corresponding to sketch data A 132a using Equation 4 above based on the first variances generated at block 505. At block 525, the example variance detector 308 generates a second weighted sum of variances (S_{yy}'). For example, the example variance detector 308 generates the second weighted sum of variances corresponding to sketch data B 132b using Equation 5 above based on the second variances generated at block 510. At block 530, the example variance detector 308 generates a weighted sum of covariances (S_{xy}'). For example, the example variance detector 308 generates the weighted sum of covariances

corresponding to sketch data A 132a and sketch data B 132b using Equation 6 above based on the covariances generated at block 515.

At block 535, the example variance detector 308 generates a first normalized weighted sum of variances (V_{xx}'). For example, the example variance detector 308 generates the first normalized weighted sum of variances corresponding to sketch data A 132a using Equation 8 above based on the first weighted sum of variances generated at block 520. At block 540, the example variance detector 308 generates a second normalized weighted sum of variances (V_{yy}'). For example, the example variance detector 308 generates the second normalized weighted sum of variances corresponding to sketch data B 132b using Equation 9 above based on the second weighted sum of variances generated at block 525. The example process of FIG. 5 then returns to a calling function or process such as implemented by the example instructions of FIG. 4.

FIG. 6 is a block diagram of an example processor platform 600 structured to execute the instructions of FIGS. 4 and 5 to implement the example audience metrics generator 112 of FIGS. 1-3. The processor platform 600 can be, for example, a server, a personal computer, a workstation, a self-learning machine (e.g., a neural network), or any other type of computing device.

The processor platform 600 of the illustrated example includes a processor 612. The processor 612 of the illustrated example is hardware. For example, the processor 612 can be implemented by one or more integrated circuits, logic circuits, microprocessors, GPUs, DSPs, or controllers from any desired family or manufacturer. The hardware processor may be a semiconductor based (e.g., silicon based) device. In this example, the processor implements the example data interface 304, the example coefficient generator 306, the example variance detector 308, the example real roots solver 310, the example data analyzer 312, the example noise checker 314, the example numerical solver 315, and the example audience size generator 316.

The processor 612 of the illustrated example includes a local memory 613 (e.g., a cache). The processor 612 of the illustrated example is in communication with a main memory including a volatile memory 614 and a non-volatile memory 616 via a bus 618. The volatile memory 614 may be implemented by Synchronous Dynamic Random Access Memory (SDRAM), Dynamic Random Access Memory (DRAM), RAMBUS® Dynamic Random Access Memory (RDRAM®) and/or any other type of random access memory device. The non-volatile memory 616 may be implemented by flash memory and/or any other desired type of memory device. Access to the main memory 614, 616 is controlled by a memory controller.

The processor platform 600 of the illustrated example also includes an interface circuit 620. The interface circuit 620 may be implemented by any type of interface standard, such as an Ethernet interface, a universal serial bus (USB), a Bluetooth® interface, a near field communication (NFC) interface, and/or a PCI express interface.

In the illustrated example, one or more input devices 622 are connected to the interface circuit 620. The input device(s) 622 permit(s) a user to enter data and/or commands into the processor 612. The input device(s) can be implemented by, for example, an audio sensor, a microphone, a camera (still or video), a keyboard, a button, a mouse, a touchscreen, a track-pad, a trackball, isopoint and/or a voice recognition system.

One or more output devices 624 are also connected to the interface circuit 620 of the illustrated example. The output

devices 624 can be implemented, for example, by display devices (e.g., a light emitting diode (LED), an organic light emitting diode (OLED), a liquid crystal display (LCD), a cathode ray tube display (CRT), an in-place switching (IPS) display, a touchscreen, etc.), a tactile output device, a printer and/or speaker. The interface circuit 620 of the illustrated example, thus, typically includes a graphics driver card, a graphics driver chip and/or a graphics driver processor.

The interface circuit 620 of the illustrated example also includes a communication device such as a transmitter, a receiver, a transceiver, a modem, a residential gateway, a wireless access point, and/or a network interface to facilitate exchange of data with external machines (e.g., computing devices of any kind) via a network 626. The communication can be via, for example, an Ethernet connection, a digital subscriber line (DSL) connection, a telephone line connection, a coaxial cable system, a satellite system, a line-of-site wireless system, a cellular telephone system, etc.

The processor platform 600 of the illustrated example also includes one or more mass storage devices 628 for storing software and/or data. Examples of such mass storage devices 628 include floppy disk drives, hard drive disks, compact disk drives, Blu-ray disk drives, redundant array of independent disks (RAID) systems, and digital versatile disk (DVD) drives.

Machine-executable instructions 632 represented by the flowchart of FIG. 4 may be stored in the mass storage device 628, in the volatile memory 614, in the non-volatile memory 616, and/or on a removable non-transitory computer readable storage medium such as a CD or DVD.

From the foregoing, it will be appreciated that example methods, apparatus and articles of manufacture have been disclosed that estimate audience sizes using deduplication based on multiple vectors of counts from multiple database proprietors using server-collected impression data. The disclosed methods, apparatus and articles of manufacture may be used to deduplicate data between any two data sets while preserving privacy. Examples of such two data sets could be audiences, email addresses, banking information, lists of purchased items, store visits, traffic patterns, etc. While examples disclosed herein are described in connection with estimating audience sizes for media, examples disclosed herein are not limited to use with audience metrics. The disclosed methods, apparatus and articles of manufacture improve the efficiency of using a computing device by reducing the amount of processing and amount of memory required to estimate unique audience sizes of media items based on audience measurements across two or more database proprietors. Using improved techniques based on multiple vectors of counts to estimate unique audience disclosed herein, the number of hashes to reach a desired confidence and accuracy declines as overlap (W) increases, whereas prior techniques for generating covariance estimates may need more and more hashes to get the same desired result. Examples disclosed herein improve the accuracy of estimating unique audience size as the number of vectors of counts provided by the database proprietors increases. The cost savings in computation, memory, and speed can be orders of magnitude in improvement over prior techniques. The disclosed methods, apparatus and articles of manufacture are accordingly directed to one or more improvement(s) in the functioning of a computer.

Example 1 includes an apparatus to determine an audience size for media based on sketch data. The apparatus of Example 1 includes a coefficient generator to determine coefficient values for a polynomial based on normalized weighted sums of variances, a normalized weighted sum of

covariances, and cardinalities corresponding to a first plurality of vectors of counts from a first database proprietor and a second plurality of vectors of counts from a second database proprietor, a real roots solver to determine a real root value of the polynomial, the real root value indicative of a number of audience members represented in the first plurality of vectors of counts that are also represented in the second plurality of vectors of counts, and an audience size generator to determine the audience size based on the real root value and the cardinalities of the first plurality of vectors of counts and the second plurality of vectors of counts.

Example 2 includes the apparatus of Example 1, and further includes a variance detector to determine the normalized weighted sums of variances and the normalized weighted sum of covariances corresponding to the first plurality of vectors of counts and the second plurality of vectors of counts.

Example 3 includes the apparatus of Example 2, wherein the variance detector is to determine the normalized weighted sums of variances and the normalized weighted sum of covariances by generating a plurality of first variances based on the first plurality of vectors of counts, generating a plurality of second variances based on the second plurality of vectors of counts, and generating a plurality of covariances based on the first plurality of vectors of counts and the second plurality of vectors of counts.

Example 4 includes the apparatus of Example 3, wherein the variance detector is to determine the normalized weighted sum of variances and the normalized weighted sum of covariances by generating a first weighted sum of variances based on the plurality of first variances, the first weighted sum of variances corresponding to a first normalized weighted sum of variances of the first plurality of vectors of counts, generating a second weighted sum of variances based on the plurality of second variances, the second weighted sum of variances corresponding to a second normalized weighted sum of variances of the second plurality of vectors of counts, and generating a weighted sum of covariances based on the plurality of covariances, the weighted sum of covariances corresponding to a normalized weighted sum of covariances of the first plurality of vectors of counts and the second plurality of vectors of counts.

Example 5 includes the apparatus of Example 4, wherein the variance detector is to generate the first weighted sum of variances, the second weighted sum of variances, and the weighted sum of covariances based on a probability factor corresponding to lengths of the first plurality of vectors of counts and the second plurality of vectors of counts.

Example 6 includes the apparatus of Example 1, and further includes a data analyzer to zero-mean data in at least one of the first plurality of vectors of counts or the second plurality of vectors of counts before the coefficient generator is to determine the coefficient values.

Example 7 includes the apparatus of Example 1, and further includes a noise checker to select equations to be used by the coefficient generator in response to obtaining a noise term from at least one of the first database proprietor or the second database proprietor.

Example 8 includes a method to determine an audience size for media based on sketch data. The method of Example 8 includes determining coefficient values for a polynomial based on normalized weighted sums of variances, a normalized weighted sum of covariances, and cardinalities corresponding to a first plurality of vectors of counts from a first database proprietor and a second plurality of vectors of counts from a second database proprietor, determining a real

root value of the polynomial, the real root value indicative of a number of audience members represented in the first plurality of vectors of counts that are also represented in the second plurality of vectors of counts, and determining the audience size based on the real root value and the cardinalities of the first plurality of vectors of counts and the second plurality of vectors of counts.

Example 9 includes the method of Example 8, and further includes determining the normalized weighted sums of variances and the normalized weighted sum of covariances corresponding to the first plurality of vectors of counts and the second plurality of vectors of counts.

Example 10 includes the method of Example 9, wherein the determining of the normalized weighted sums of variances and the normalized weighted sum of covariances includes generating a plurality of first variances based on the first plurality of vectors of counts, generating a plurality of second variances based on the second plurality of vectors of counts, and generating a plurality of covariances based on the first plurality of vectors of counts and the second plurality of vectors of counts.

Example 11 includes the method of Example 10, wherein the determining of the normalized weighted sum of variances and the normalized weighted sum of covariances includes generating a first weighted sum of variances based on the plurality of first variances, the first weighted sum of variances corresponding to a first normalized weighted sum of variances of the first plurality of vectors of counts, generating a second weighted sum of variances based on the plurality of second variances, the second weighted sum of variances corresponding to a second normalized weighted sum of variances of the second plurality of vectors of counts, and generating a weighted sum of covariances based on the plurality of covariances, the weighted sum of covariances corresponding to a normalized weighted sum of covariances of the first plurality of vectors of counts and the second plurality of vectors of counts.

Example 12 includes the method of Example 11, wherein the generating of the first weighted sum of variances, the second weighted sum of variances, and the weighted sum of covariances is based on a probability factor corresponding to lengths of the first plurality of vectors of counts and the second plurality of vectors of counts.

Example 13 includes the method of Example 8, and further includes zero-meaning data in at least one of the first plurality of vectors of counts or the second plurality of vectors of counts before determining the coefficient values.

Example 14 includes the method of Example 8, and further includes selecting equations to be used in response to obtaining a noise term from at least one of the first database proprietor or the second database proprietor.

Example 15 includes a non-transitory computer readable storage medium to determine an audience size for media based on sketch data. The non-transitory computer readable storage medium of Example 15 includes instructions that, when executed, cause at least one processor to at least determine coefficient values for a polynomial based on normalized weighted sums of variances, a normalized weighted sum of covariances, and cardinalities corresponding to a first plurality of vectors of counts from a first database proprietor and a second plurality of vectors of counts from a second database proprietor, determine a real root value of the polynomial, the real root value indicative of a number of audience members represented in the first plurality of vectors of counts that are also represented in the second plurality of vectors of counts, and determine the audience size based on the real root value and the cardinali-

ties of the first plurality of vectors of counts and the second plurality of vectors of counts.

Example 16 includes the non-transitory computer readable storage medium of Example 15, wherein the instructions, when executed, cause the processor to determine the normalized weighted sums of variances and the normalized weighted sum of covariances corresponding to the first plurality of vectors of counts and the second plurality of vectors of counts.

Example 17 includes the non-transitory computer readable storage medium of Example 16, wherein the instructions, when executed, cause the processor to determine the normalized weighted sums of variances and the normalized weighted sum of covariances by generating a plurality of first variances based on the first plurality of vectors of counts, generating a plurality of second variances based on the second plurality of vectors of counts, and generating a plurality of covariances based on the first plurality of vectors of counts and the second plurality of vectors of counts.

Example 18 includes the non-transitory computer readable storage medium of Example 17, wherein the instructions, when executed, cause the processor to determine the normalized weighted sum of variances and the normalized weighted sum of covariances by generating a first weighted sum of variances based on the plurality of first variances, the first weighted sum of variances corresponding to a first normalized weighted sum of variances of the first plurality of vectors of counts, generating a second weighted sum of variances based on the plurality of second variances, the second weighted sum of variances corresponding to a second normalized weighted sum of variances of the second plurality of vectors of counts, and generating a weighted sum of covariances based on the plurality of covariances, the weighted sum of covariances corresponding to a normalized weighted sum of covariances of the first plurality of vectors of counts and the second plurality of vectors of counts.

Example 19 includes the non-transitory computer readable storage medium of Example 18, wherein the instructions, when executed, cause the processor to generate the first weighted sum of variances, the second weighted sum of variances, and the weighted sum of covariances based on a probability factor corresponding to lengths of the first plurality of vectors of counts and the second plurality of vectors of counts.

Example 20 includes the non-transitory computer readable storage medium of Example 15, wherein the instructions, when executed, cause the processor to zero-mean data in at least one of the first plurality of vectors of counts or the second plurality of vectors of counts before determining the coefficient values.

Example 21 includes the non-transitory computer readable storage medium of Example 15, wherein the instructions, when executed, cause the processor to select equations to be used in response to obtaining a noise term from at least one of the first database proprietor or the second database proprietor.

Example 22 includes an apparatus to determine an audience size for media based on sketch data. The apparatus of Example 22 includes means for determining coefficient values for a polynomial based on normalized weighted sums of variances, a normalized weighted sum of covariances, and cardinalities corresponding to a first plurality of vectors of counts from a first database proprietor and a second plurality of vectors of counts from a second database proprietor, means for determining a real root value of the polynomial, the real root value indicative of a number of audience members represented in the first plurality of vectors of

counts that are also represented in the second plurality of vectors of counts, and means for determining the audience size based on the real root value and the cardinalities of the first plurality of vectors of counts and the second plurality of vectors of counts.

Example 23 includes the apparatus of Example 22, and further includes means for determining the normalized weighted sums of variances and the normalized weighted sum of covariances corresponding to the first plurality of vectors of counts and the second plurality of vectors of counts.

Example 24 includes the apparatus of Example 23, wherein the means for determining the normalized weighted sums of variances and the normalized weighted sum of covariances is to determine the normalized weighted sums of variances and the normalized weighted sum of covariances by generating a plurality of first variances based on the first plurality of vectors of counts, generating a plurality of second variances based on the second plurality of vectors of counts, and generating a plurality of covariances based on the first plurality of vectors of counts and the second plurality of vectors of counts.

Example 25 includes the apparatus of Example 24, wherein the means for determining the normalized weighted sums of variances and the normalized weighted sum of covariances is to determine the normalized weighted sum of variances and the normalized weighted sum of covariances by generating a first weighted sum of variances based on the plurality of first variances, the first weighted sum of variances corresponding to a first normalized weighted sum of variances of the first plurality of vectors of counts, generating a second weighted sum of variances based on the plurality of second variances, the second weighted sum of variances corresponding to a second normalized weighted sum of variances of the second plurality of vectors of counts, and generating a weighted sum of covariances based on the plurality of covariances, the weighted sum of covariances corresponding to a normalized weighted sum of covariances of the first plurality of vectors of counts and the second plurality of vectors of counts.

Example 26 includes the apparatus of Example 25, wherein the means for determining the normalized weighted sums of variances and the normalized weighted sum of covariances is to generate the first weighted sum of variances, the second weighted sum of variances, and the weighted sum of covariances based on a probability factor corresponding to lengths of the first plurality of vectors of counts and the second plurality of vectors of counts.

Example 27 includes the apparatus of Example 22, and further includes means for zero-meaning data in at least one of the first plurality of vectors of counts or the second plurality of vectors of counts before the means for determining the coefficient values is to determine the coefficient values.

Example 28 includes the apparatus of Example 22, and further includes means for selecting equations to be used by the means for determining the coefficient values in response to obtaining a noise term from at least one of the first database proprietor or the second database proprietor.

Although certain example methods, apparatus and articles of manufacture have been disclosed herein, the scope of coverage of this patent is not limited thereto. On the contrary, this patent covers all methods, apparatus and articles of manufacture fairly falling within the scope of the claims of this patent.

The invention claimed is:

1. A computing system comprising a processor and a memory, the computing system configured to perform a set of operations comprising:

obtaining first sketch data representing a first audience of
a media item, wherein the first audience includes first
subscribers of a first database proprietor, wherein the
first sketch data comprises a first plurality of vectors of
counts, wherein each vector of counts of the first
plurality of vectors of counts represents the first audi-
ence as a distribution of first hash values, wherein each
vector of counts of the first plurality of vectors of
counts includes a plurality of bins having respective bin
numbers, wherein the first hash values are usable to
determine bin numbers for assigning the first subscri-
bers to respective bins of the plurality of bins, and
wherein the first hash values are generated using a hash
algorithm;

obtaining second sketch data representing a second audi-
ence of the media item, wherein the second sketch data
comprises a second plurality of vectors of counts,
wherein each vector of counts of the second plurality of
vectors of counts represents the second audience as a
distribution of second hash values, wherein each vector
of counts of the second plurality of vectors of counts
includes a plurality of bins having respective bin num-
bers, and wherein the second hash values are usable to
determine bin numbers for assigning audience mem-
bers of the second audience to respective bins of the
plurality of bins;

determining coefficient values for a polynomial based on
(a) normalized weighted sums of variances, (b) a nor-
malized weighted sum of covariances, and (c) cardin-
alities, the normalized weighted sums of variances,
the normalized weighted sum of covariances, and the
cardinalities corresponding to the first plurality of vec-
tors of counts and the second plurality of vectors of
counts;

determining a real root value of the polynomial, the real
root value indicative of a number of audience members
of the first audience represented in the first plurality of
vectors of counts that are also represented in the second
plurality of vectors of counts;

determining an audience size based on the real root value
and the cardinalities of the first plurality of vectors of
counts and the second plurality of vectors of counts;
and

outputting the audience size.

2. The computing system of claim 1, wherein obtaining
the first sketch data comprises obtaining the first sketch data
from the first database proprietor via a network communi-
cation from a server of the first database proprietor.

3. The computing system of claim 1, wherein the second
hash values are generated using the hash algorithm.

4. The computing system of claim 1, wherein outputting
the audience size comprises outputting the audience size to
another computing system.

5. The computing system of claim 1, wherein the second
audience includes second subscribers of a second database
proprietor.

6. A method comprising:

obtaining, by a computing system, first sketch data rep-
resenting a first audience of a media item, wherein the
first audience includes first subscribers of a first data-
base proprietor, wherein the first sketch data comprises
a first plurality of vectors of counts, wherein each
vector of counts of the first plurality of vectors of

counts represents the first audience as a distribution of
first hash values, wherein each vector of counts of the
first plurality of vectors of counts includes a plurality of
bins having respective bin numbers, wherein the first
hash values are usable to determine bin numbers for
assigning the first subscribers to respective bins of the
plurality of bins, and wherein the first hash values are
generated using a hash algorithm;

obtaining, by the computing system, second sketch data
representing a second audience of the media item,
wherein the second sketch data comprises a second
plurality of vectors of counts, wherein each vector of
counts of the second plurality of vectors of counts
represents the second audience as a distribution of
second hash values, wherein each vector of counts of
the second plurality of vectors of counts includes a
plurality of bins having respective bin numbers, and
wherein the second hash values are usable to determine
bin numbers for assigning audience members of the
second audience to respective bins of the plurality of
bins;

determining, by the computing system, coefficient values
for a polynomial based on (a) normalized weighted
sums of variances, (b) a normalized weighted sum of
covariances, and (c) cardinalities, the normalized
weighted sums of variances, the normalized weighted
sum of covariances, and the cardinalities corresponding
to the first plurality of vectors of counts and the second
plurality of vectors of counts;

determining, by the computing system, a real root value of
the polynomial, the real root value indicative of a
number of audience members of the first audience
represented in the first plurality of vectors of counts
that are also represented in the second plurality of
vectors of counts;

determining, by the computing system, an audience size
based on the real root value and the cardinalities of the
first plurality of vectors of counts and the second
plurality of vectors of counts; and

outputting the audience size.

7. The method of claim 6, wherein obtaining the first
sketch data comprises obtaining the first sketch data from
the first database proprietor via a network communication
from a server of the first database proprietor.

8. The method of claim 6, wherein the second hash values
are generated using the hash algorithm.

9. The method of claim 6, wherein outputting the audience
size comprises outputting the audience size to another
computing system.

10. The method of claim 6, wherein the second audience
includes second subscribers of a second database proprietor.

11. A non-transitory computer-readable storage medium
having stored therein instructions that, upon execution by a
computing system, cause the computing system to perform
a set of operations comprising:

obtaining first sketch data representing a first audience of
a media item, wherein the first audience includes first
subscribers of a first database proprietor, wherein the
first sketch data comprises a first plurality of vectors of
counts, wherein each vector of counts of the first
plurality of vectors of counts represents the first audi-
ence as a distribution of first hash values, wherein each
vector of counts of the first plurality of vectors of
counts includes a plurality of bins having respective bin
numbers, wherein the first hash values are usable to
determine bin numbers for assigning the first subscrib-

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ers to respective bins of the plurality of bins, and wherein the first hash values are generated using a hash algorithm;

obtaining second sketch data representing a second audience of the media item, wherein the second sketch data comprises a second plurality of vectors of counts, wherein each vector of counts of the second plurality of vectors of counts represents the second audience as a distribution of second hash values, wherein each vector of counts of the second plurality of vectors of counts includes a plurality of bins having respective bin numbers, and wherein the second hash values are usable to determine bin numbers for assigning audience members of the second audience to respective bins of the plurality of bins;

determining coefficient values for a polynomial based on (a) normalized weighted sums of variances, (b) a normalized weighted sum of covariances, and (c) cardinalities, the normalized weighted sums of variances, the normalized weighted sum of covariances, and the cardinalities corresponding to the first plurality of vectors of counts and the second plurality of vectors of counts;

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determining a real root value of the polynomial, the real root value indicative of a number of audience members of the first audience represented in the first plurality of vectors of counts that are also represented in the second plurality of vectors of counts;

determining an audience size based on the real root value and the cardinalities of the first plurality of vectors of counts and the second plurality of vectors of counts; and

outputting the audience size.

12. The non-transitory computer-readable storage medium of claim **11**, wherein obtaining the first sketch data comprises obtaining the first sketch data from the first database proprietor via a network communication from a server of the first database proprietor.

13. The non-transitory computer-readable storage medium of claim **11**, wherein the second hash values are generated using the hash algorithm.

14. The non-transitory computer-readable storage medium of claim **11**, wherein outputting the audience size comprises outputting the audience size to another computing system.

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